

What Remains of Cross-Country Convergence?[†]

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We examine the record of cross-country growth over the past fifty years and ask if developing countries have made progress on closing the income gap between their per capita incomes and those in the advanced economies. We conclude that, as a group, they have not and then survey the literature on absolute convergence with particular emphasis on that from the last decade or so. That literature supports our conclusion of a lack of progress in closing the income gap between countries. We close with a brief examination of the recent literature on cross-individual distribution of income, which finds that despite the lack of progress on cross country convergence, global inequality has tended to fall since 2000. (JEL E01, E13, O11, O47, F41, F62)

1. Introduction

Few topics in applied economic research have received as much attention, over the last three decades, as the convergence hypothesis. The hypothesis in its simplest form states that initial conditions have no implications for a country's per capita income level in the long run. In practice, the

hypothesis is often taken to mean that per capita incomes in different countries are getting closer to each other in some sense, which implies that poorer countries are catching up with richer countries. Conceptually scrutinizing and empirically testing this hypothesis became prominent with the emergence of modern growth theory in the mid-1980s, as tackling this hypothesis was seen as an important part of the project of unlocking the mechanics of economic growth; as Lucas (1988) eloquently argued in his pioneering article, “The consequences for human welfare involved in questions like these are simply staggering.”

This review aims at providing a critical assessment of this substantive but elusive concept that has generated an extensive literature that is at the heart of growth theory, empirics, and policy. What makes an old concept once again topical is the changing landscape of the global economy, the

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heterogeneity of growth experiences only recently being unraveled, and the emergence of superpowers that unexpectedly challenged what we think is possible and under what conditions.

Given the vast literature on this topic, it is important to clarify at the outset what this review does and does not do. While key ideas from previous work will have to be explained again, this review will place more emphasis on work done in the last ten to fifteen years, or that was missed, omitted, or given less emphasis by earlier surveys. For example, in terms of conceptual framework, we aim to emphasize papers that have a tighter connection between theory and evidence than has been the norm in the literature, or where predictions about convergence go beyond the neoclassical model. It should also be stressed here that while regional convergence is a large area in itself, the emphasis of this survey is squarely on international income convergence. Also, while examining convergence in other key macro variables, such as welfare and health, is as interesting and important, our attention here is on per capita income convergence as we do not want to risk a further loss of focus.

According to Maddison (2007), the industrial revolution and colonialism brought about great divergence in incomes across countries. Milanovic (2012) estimates that between the early 1800s and 1950, the average per capita income gap between industrial and less developed countries rose from a factor of 3 or 4 to a factor of 20 or more. This divergence slowed after World War II, which also coincided with the end of colonialism. Since the early 1990s, the pace of growth of income per capita in many developing economies has accelerated to unprecedented levels and is substantially above that in high-income countries. This wave of potential catching-up behavior coincides with another key global fact: namely that in many of the same countries that have

experienced rapid growth, the distribution of income has become more unequal because the share of top earners' income has often risen dramatically. The same period has seen dismal growth in a group of very poor fragile states that have been unable to participate in the recent wave of country growth described above, owing to wars and political unrest. It is unclear exactly how these developments have impacted the global distribution of income across individuals. Pinkovskiy and Sala-i-Martin (2009, 2014), for example, argue that world Gini coefficients and poverty rates have fallen since the mid-1970s, while others such as Chen and Ravallion (2010) find less evidence of poverty reduction and argue that there are important regional differences in the extent of progress.¹

The proposition that initial conditions have no implications for the long-run distribution of per capita income is motivated by the single stable steady state of the neoclassical model with a globally diminishing marginal product of capital.² By contrast, models with constant or increasing returns can exhibit a multiplicity or absence of stable steady states.³ In the latter case, a country's long-run distribution of per capita income can depend on its initial conditions; for example, a model with a feedback loop from income to population growth that has a high population growth–low income steady state and a low population growth–high income steady state with countries being selected to move to one steady state or the other based on their initial level of per capita income. Against this background, a statistical test

¹ Anand, Segal, and Stiglitz (2010) contains several analyses of the measurement issues involved and the conclusions of this debate.

² Strictly speaking, something like the Inada (1963) conditions are needed to guarantee a unique stable steady state in this model.

³ Azariadis (1996) and Galor (1996) present surveys of theoretical mechanisms that can produce a multiplicity of steady states.

with a null hypothesis of non-convergence and an alternative hypothesis of convergence can be seen as potentially informative about the sorts of models best suited to the study of economic growth.

The convergence issue has important policy implications because, if there is a single stable steady state, small-scale policy interventions can be helpful to the extent that they hasten the transition of the poor countries to inevitable prosperity. However, if there are multiple stable steady states, then large-scale policy interventions may be required to push poor economies from one basin of attraction to another. In such a world, small-scale policy interventions that fail to push countries into a different basin of attraction will eventually be undone as economies fall back to the steady state associated with the basin of attraction in which they lie. In broad terms, a multiplicity of steady states can lead to the sorts of poverty traps discussed by Collier (2007) or the hypothesized middle-income trap studied by Eichengreen, Park, and Shin (2014).

The plan for this review is as follows: We start section 2 by demonstrating the key facts and patterns of per capita GDP growth across countries over the last fifty years. We consider not only differences in growth across countries but also differences over time, with a view to getting a sense of the varied evolution of the per capita incomes in different subgroups of countries. In section 3 we first consider the definition and operationalization of the convergence concept, before discussing the theoretical foundations of the catching-up process in the closed-economy neoclassical growth model and in a more realistic open-economy model. Having examined different concepts and theoretical models of convergence, in section 4 we provide a brief summary of the tests employed in the empirical growth literature to estimate convergence. In section 5 we briefly touch on econometric issues facing estimation of

convergence, before turning to empirical evidence in section 6. Given how vast the literature is on the subject, we chose to focus attention to only a selected set of key papers from panel, time series, and distributional approaches. Section 7 concludes with our assessment of the evidence and a discussion on the lessons learned so far from this voluminous literature.

2. *Patterns and Facts on Growth over the Past Half Century*

2.1 *Context*

This section presents what are, in our view, some of the more important growth patterns and facts from the past half century. We concentrate on reporting differences in growth across countries but also differences over time. The objective is to contextualize, in a very intuitive way, patterns of convergence, divergence, and other interesting growth dynamics using some very basic statistics and figures.

While it would be unreasonable to claim that fifty years' worth of data are nearly enough to draw out the main facts about long-run growth, we believe that these data can adequately and meaningfully reflect modern facts on growth.⁴ In this short period of time, by historical standards, the world has seen unprecedented economic progress that has spanned the globe. The past half century has witnessed large improvements in living standards across many parts of the world that have transformed the prevailing landscape of economic thinking and policy. From this perspective, we aim to exposit some of the key growth trends that emerged over the past half century as parsimoniously as possible.

The analysis below is based on the Penn World Table (PWT) version 7.1 database,

⁴The identification problem discussed by Durlauf, Johnson, and Temple (2005, pp. 622–23) is relevant here.

covering 182 countries over the period 1950–2010.⁵ This data set has been the gold standard of cross-country data sets and has been used extensively in the empirical growth literature, including tests of convergence—roughly 70 percent of cross-country empirical work is based on PWT, followed by World Bank’s World Development Indicators (WDI), and the IMF’s World Economic Outlook (WEO) data set placed a distant third. This is largely because the PWT has managed to curb well-known measurement issues related to cross-country comparability through more than forty years of continuous improvements in data collection and methodologies.⁶

2.2 *Global Growth*

The world as a whole has experienced unprecedented economic growth over the last half century. Average per capita purchasing power parity (PPP) adjusted GDP across the globe increased from US\$4,155 in 1960 to US\$13,368 in 2014 implying an average annual growth of 4 percent (see

figure 1).⁷ While hard evidence to corroborate this claim is at best patchy, there exists a consensus among social scientists that the past half century has been the most prosperous in terms of world economic growth and welfare. At the same time, growth has been uneven across countries and also across different time periods. We consider this heterogeneity as a defining feature of the modern growth experience as we take a closer look at country experiences across income and time, next.

2.3 *Heterogeneity across Geographical Regions*

Table 1 reports average decadal growth rates in seven geographical regions as commonly classified by the IMF and the World Bank (East Asia and Pacific, Europe and Central Asia, Latin America and Caribbean, Middle East and North Africa, North America, South Asia, and sub-Saharan Africa) during the period 1960–2010. Global growth was relatively stable in the 1960s and 1970s, with all regions having positive average growth rates in the range of about 3–4 percent with the exception of sub-Saharan Africa and South Asia, which grew below 2 percent per year.

Growth rates in all regions experienced a significant decline in the 1980s, with the exception of South Asia, resulting in world growth taking a large dip. Latin America, Middle East and North Africa, and sub-Saharan Africa experienced the greatest declines, with growth falling into negative territory. For these regions the 1980s was a “lost decade,” but with the exception of sub-Saharan Africa, these regions began to recover in the 1990s and 2000s.

South Asia followed a different pattern; it continued the growth of the 1960s and 1970s

⁵The main contribution of the PWT was to convert national measures of GDP and income into internationally comparable PPP estimates. This is done by collecting prices for the same or similar goods in different countries and deriving price indices that can be used to compare what people can actually buy. The massive undertaking of price collection, known as the International Comparison Programme/Project (ICP) is at the heart of PWT—see Deaton and Heston (2010) and Johnson et al. (2013) for detailed discussions on methodological challenges facing PWT. The PWT has reported on eight rounds of data, starting in 1970 (for more details see the pioneering papers by Kravis, Heston, and Summers 1978, and Summers and Heston 1988). There are other data sets we could use, most notably Angus Maddison’s historical GDP data, but unfortunately that data set, as extensive as it is in the time dimension, it only covers a limited number of developing countries.

⁶For the stylized facts presented in this section we consider countries with populations above 1 million, to exclude small states that are likely to follow unique growth experiences. The time horizon of our analysis below is 1960–2010 as the 1950s data is quite fragmented and mostly available for industrialized economies. Using the latest published version of the tables (PWT 9.0 released in August 2016) does not change our assessment of cross-country trends.

⁷Global growth rates were calculated by taking the mean of real PPP adjusted per capita GDP growth across countries.

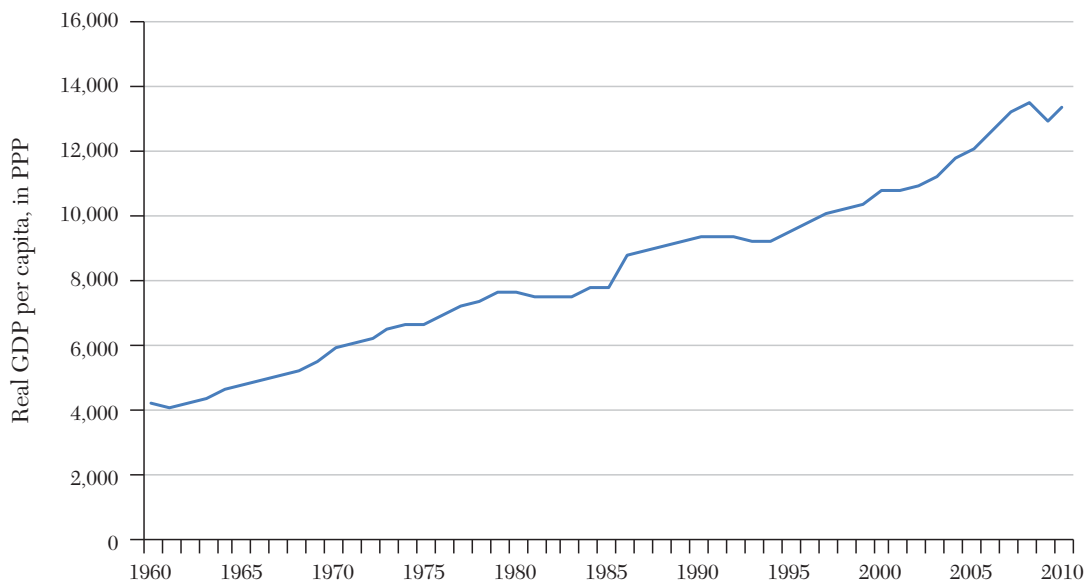


Figure 1. Average Global per Capita GDP (1960–2010)

Source: Penn World Tables version 7.1.

TABLE 1
DECADAL AVERAGE PER CAPITA GDP GROWTH (%) BY GEOGRAPHICAL REGION

Geographical Region	1960s	1970s	1980s	1990s	2000s
East Asia and Pacific	3.9	3.3	3.2	3.0	3.6
Europe and Central Asia	4.7	3.5	1.8	0.5	3.6
Latin America and Caribbean	2.2	2.7	−0.6	1.5	2.2
Middle East and North Africa	3.7	2.7	−0.9	2.0	2.0
North America	3.1	2.5	1.9	1.8	0.9
South Asia	1.6	1.4	2.1	0.5	4.5
Sub-Saharan Africa	1.8	1.3	−0.2	−0.4	1.8
World	2.8	2.4	0.6	0.9	2.7

Source: Penn World Tables version 7.1.

into the 1980s but saw decelerating growth in the 1990s. This was followed by a dramatic bounce back in the 2000s, during which it was the fastest growing region, recording an annual growth of 4.5 percent, something

only seen previously in the 1960s in Europe. Notable also is sub-Saharan Africa's bounce back in the 2000s after very poor performances in the previous two decades (see Johnson, Ostry, and Subramanian 2007).

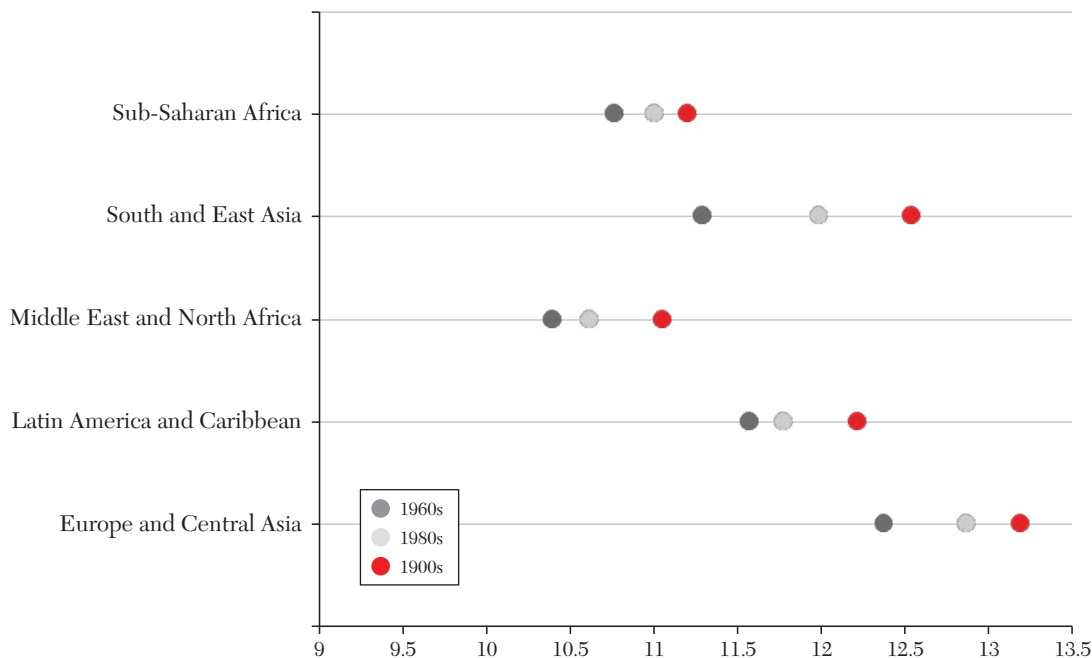


Figure 2a. Log of per Capita Income by Region (1960–2010)

Source: Penn World Tables version 7.1 based on balanced sample of countries. Countries with population below 1 million were dropped from the sample.

The strong performance of some African countries (such as Rwanda, Tanzania, and Botswana) had generated sentiments of optimism by some economists (see Miguel 2009, Radelet 2010), while others flagged the sharp rise in commodity prices as the main driver of this experience. Figure 2a provides an illustration of regional progress from 1960 to 2010 with South and East Asia showing remarkable improvements in per capita GDP while sub-Saharan Africa fell behind (see also figure 3).

2.4 Heterogeneity across Income Groups and Exporter Type

Table 2 presents average growth in three income groups—high-income countries (HICs), middle-income countries (MICs), and low-income countries (LICs)—along

with a group of countries classified as commodity exporters.⁸ The data reveal that the three income groups followed distinct growth patterns in the past half century.

HICs began as the fastest growing set of economies, with an average annual growth of 4.7 percent in the 1960s, but have gradually become the slowest growing group in the 2000s, at 1.7 percent annual growth. In contrast, both MICs and LICs saw growth deteriorate drastically in the 1980s, before experiencing a recovery in the 2000s. Specifically, MICs suffered a severe reduction of average annual growth, from around 3 percent in the 1960s and 1970s to a dismal 0.4 percent in the 1980s. They have since

⁸See online appendix table A.1 for a list of countries in each grouping.

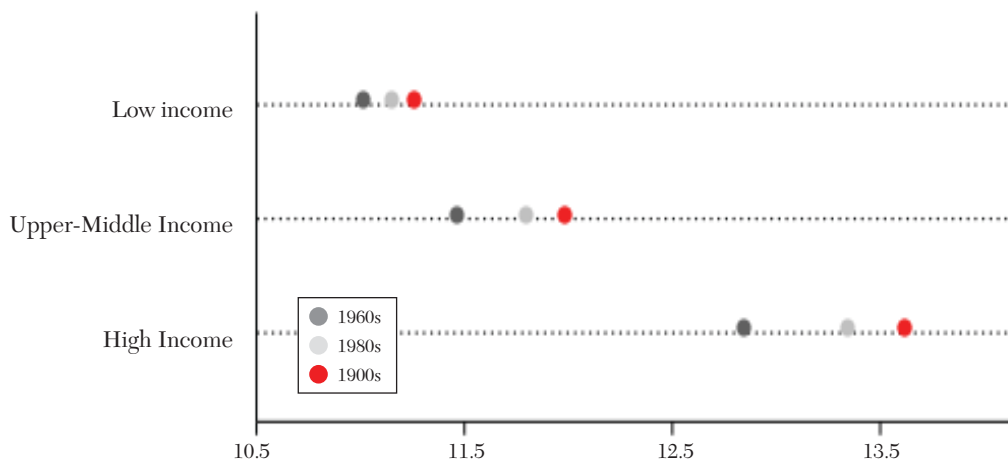


Figure 2b. Log of per Capita Income by Region (1960–2010)

Source: Penn World Tables version 7.1 based on balanced sample of countries. Countries with population below 1 million were dropped from the sample.

then recorded a strong recovery reaching an average of 3.4 percent in the 2000s.

LICs, on the other hand, faced continuous decline every decade from the 1960s to the 1990s, with negative growth rates in the 1980s and 1990s, before the surprising and unprecedented resurgence in the 2000s with 2.4 percent growth. Given recent growth successes, some LICs went from countries associated with disaster in the pre-2000 era to success stories in the post-2000 era. While many policy makers and commentators have hailed LICs' recent achievements, researchers are still trying to understand where this newly found growth is coming from. More broadly, during the 2000s we observe some convergence in relative income levels as MICs and LICs showed considerably higher growth rates than HICs. Recent data unfortunately cast doubt as to whether this upward growth trend will continue, as growth in many commodity exporting LICs has started to decelerate as commodity prices declined sharply.

The bottom panel of table 2 reports average growth rates in each of the decades in our sample for a commodity-exporting group of countries including oil and precious mineral exporters. These countries had negative growth in the 1980s and 1990s, but saw a marked reversal of fortunes in the 2000s with commodity prices exploding during the same period—for example crude oil prices shot up from around US\$19 per barrel in 2000 to over US\$130 in the first quarter of 2008 before plummeting to under US\$40 in 2016; similarly, the price of gold climbed from US\$274 per ounce in 2000 to a record high of US\$1,405 in 2010).

Figure 2b illustrates the way that HICs and MICs have opened the gap that separates them from income levels in LICs. In addition, figure 3, which plots per capita GDP in a few selected countries and income groups relative to that in the United States, paints a similar picture.

Figure 4 plots the growth rate over 1960–2010 against the 1960 level of real per

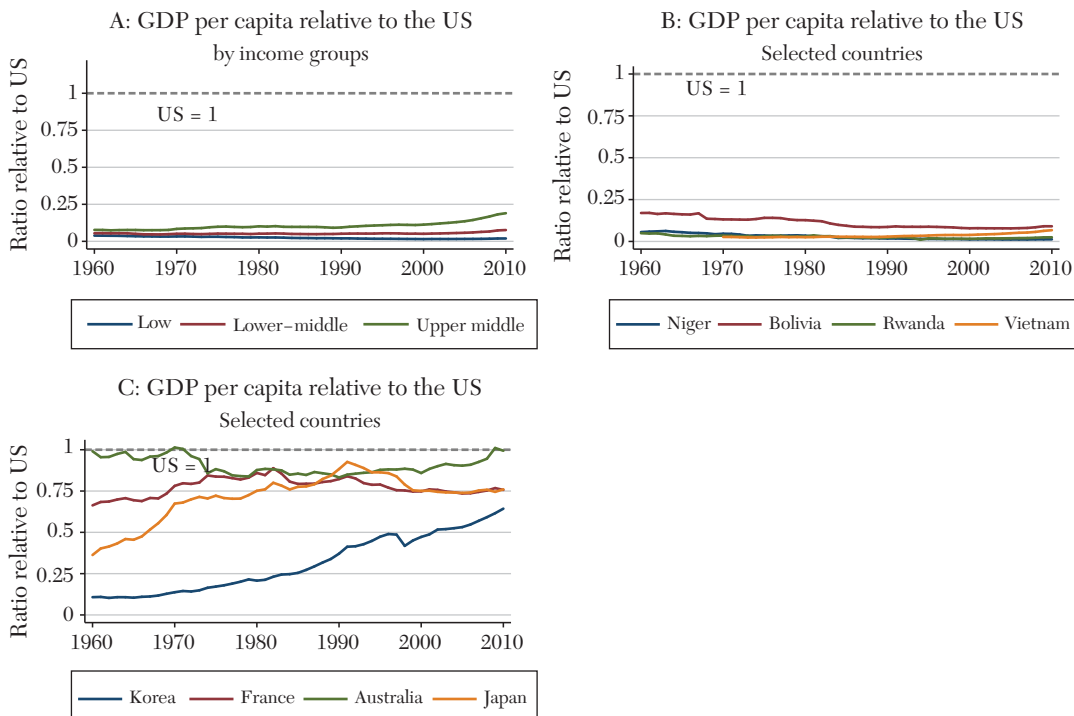


Figure 3. Income Levels Relative to the United States (1960–2010)

Notes: Income definition based on PPP converted GDP per capita (chain series), at 2005 constant prices. Countries with incomplete data were dropped out from calculations. Countries with population less than 1 million were dropped out from sample.

Source: Penn World Tables 7.1.

TABLE 2
DECADAL AVERAGE PER CAPITA GDP GROWTH (%) BY INCOME AND EXPORTER GROUPS

	1960s	1970s	1980s	1990s	2000s
<i>Income Group</i>					
HIC	4.7	3.3	2.4	2.1	1.7
MIC	2.8	3.4	0.4	1.4	3.4
LIC (all)	1.4	0.7	−0.2	−0.5	2.4
LIC (fragile)	1.7	0.7	−0.5	−1.5	1.3
LIC (non-fragile)	1.1	0.7	0.2	0.6	3.6
<i>Exporter Group</i>					
Commodity Exporters	2.1	2.0	−0.8	−0.4	3.0
Others	3.0	2.5	1.1	1.3	2.7
World	2.8	2.4	0.6	0.9	2.7

Source: Penn World Tables version 7.1.

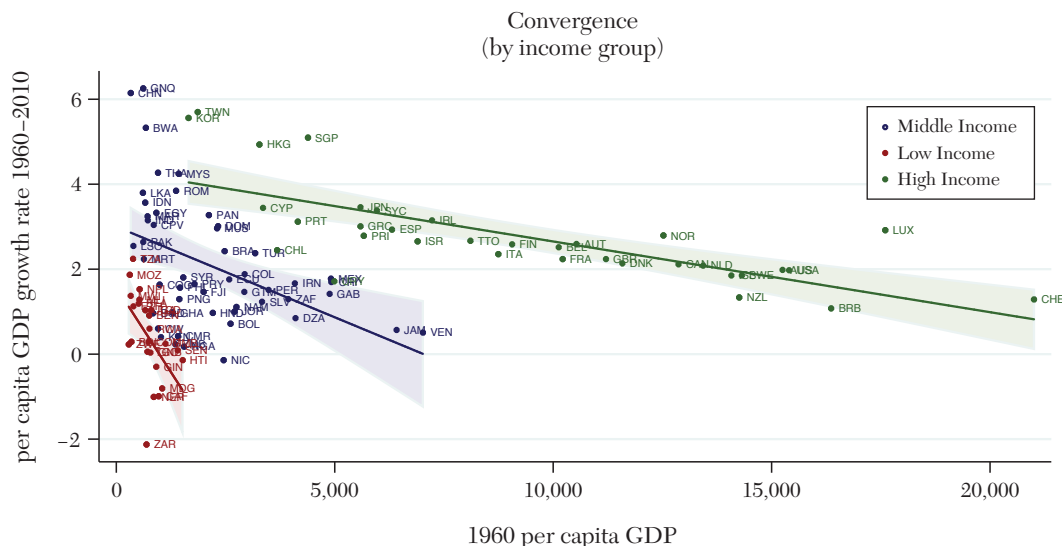


Figure 4. Growth Against Initial Income

Notes: Income definition based on PPP converted GDP per capita (chain series), at 2005 constant prices. Fitted values are shown for each group.

Source: Penn World Tables 7.1.

capita GDP for all countries for which data exist in PWT 7.1. Different versions of this plot, which present one of the most well-documented features of cross-country growth experience, have appeared in many books and papers. Some interesting observations are immediately apparent. First, countries early in their development process exhibit much more diverse growth experiences over the period compared to more advanced countries. Second, once LICs, MICs and HICs are indicated in different colors, catching-up behavior, in the form of a negative relationship between initial income and subsequent growth, is evident within each of the groups but not across the all countries. That is, local but not global catching-up behavior is observed.⁹

⁹It is important to be mindful of De Long's (1988) criticism of Baumol's (1986) early observation of the different growth behavior in LICs, MICs, and HICs on the basis of sample selection and measurement error bias.

Figures 5 and 6 present estimated cross-country per capita income distributions for 1960 and 2010 and standard deviations for the distribution for 1960–2010, respectively. Calculations in both figures used a balance sample of 110 countries for which data were available. Notwithstanding the crude nature of these two figures, they reveal some notable patterns. The good news from figure 5 is that the median country income improved markedly between 1960 and 2010, from US\$2,161 to US\$6,682 (in constant prices, a growth rate of 2.3 percent per year), as indicated by the rightward shift of the center of the distribution. In 2010, only 30 percent of the countries had per capita incomes below the 1960 median, providing some evidence of the transition of some low- and middle-income countries to higher-income status. However, the figure also contains bad news inasmuch as the distribution of per capita income in countries around

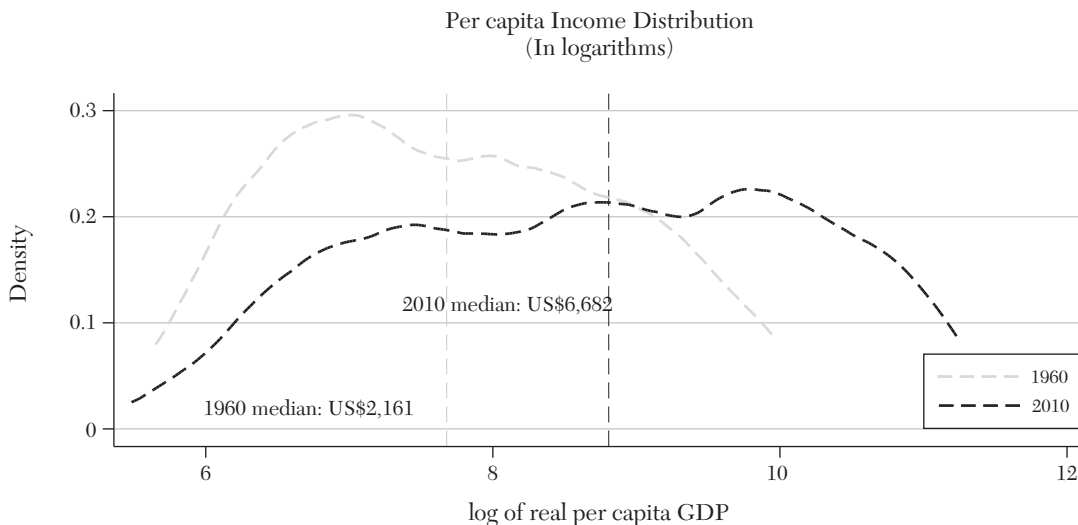


Figure 5. Cross-country Income Distribution against log per Capita GPD (1960, 2010)

Notes: Income definition based on PPP converted GDP per capita (chain series), at 2005 constant prices. Vertical lines denote median of values for respective year. Sample of countries constant across years. The densities shown are standard kernel density estimators calculated using the Epanechnikov kernel.

Source: Penn World Tables 7.1.

the world has become more disbursed over this period as the distribution elongated and became flatter.

Figure 6 shows a continuous increase in the cross-country standard deviations from 1960 until the mid-1990s, followed by a notable slowing and reversal of this trend in the mid-to late 2000s. The implied cessation of the rise in the cross-country dispersion of per capita income sits well with the narrative regarding the remarkable growth spurt experienced by many emerging market and developing economies since the mid-1990s—a narrative also consistent with the 2010 cross-country income distribution depicted in figure 5.

2.5 Heterogeneity within LICs

Table 2 also makes a distinction between what are commonly called fragile and non-fragile LICs, with the former comprising

about one-third of the LICs. Fragile states are defined as countries facing political fragility, characterized by weak institutional capacity, poor governance, and conflict.¹⁰ It is important to make this distinction, as there is significant variation in the experience between fragile states and the rest of LICs especially in the last two decades of our sample period. Concretely, there is a difference of over 2 percent annual growth between the two groups in both decades. This relatively recent development is quite important because it highlights that while there is a lot of optimism over the most recent growth acceleration in LICs, aggregating their experience masks a large

¹⁰We use the World Bank's definition of fragility (see online appendix table A.1 for the list of countries which formed the baseline sample of fragile states in the analysis). On a cautionary note, we are agnostic as to whether fragile states are growing less due to their fragility or causality might go the other way around.

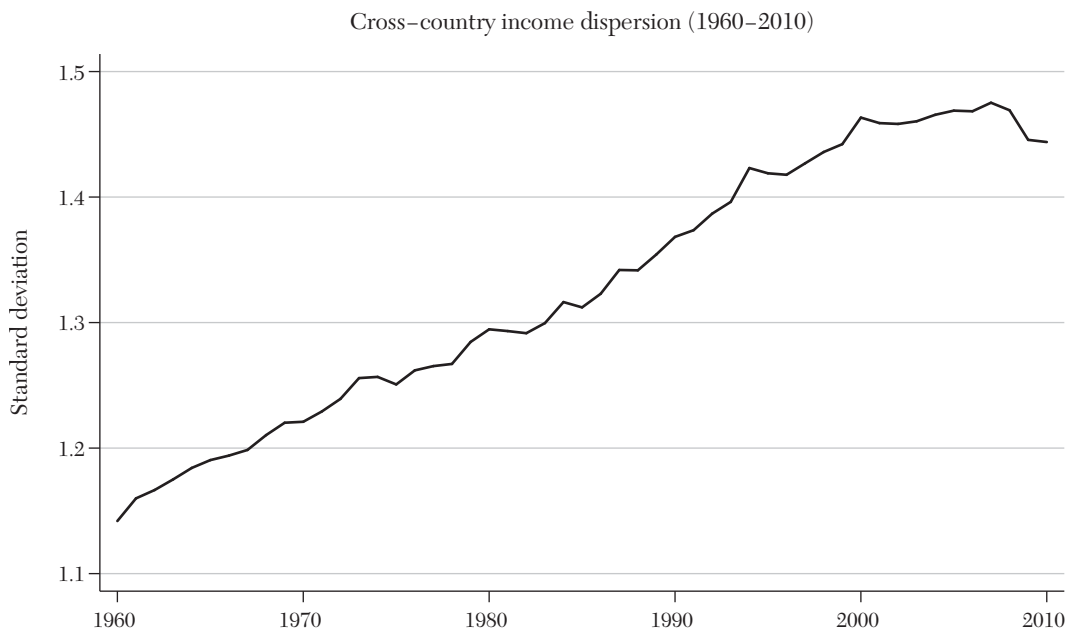


Figure 6. Cross-country Income Dispersion (1960–2010)

Notes: Income definition based on PPP converted GDP per capita (chain series), at 2005 constant prices. Sample of countries constant across years.

Source: Penn World Tables 7.1.

differential the performance of fragile and non-fragile states. The story that emerges is one of divergence within the LIC group. Put differently, the unprecedented growth acceleration observed in the LIC group as a whole over the past decade, after a long period of poor performance, masks the fact that only about half of these countries are contributing to the resurgence while the rest of the countries are stagnant.

2.6 Top and Bottom Growth Performers across Time

Table 3 reports the top ten and bottom ten growth performers for each decade in our sample. There are a number of observations that are worth making using these

data. First, it is quite stunning how China moved from one of the ten worst growth performers in the 1960s (with -0.32 percent annual growth rate) to claim the first place in the list of top performers in the 1990s and 2000s (see Subramanian 2011 for an articulate and convincing account of China's rise to economic prominence). The growth experience of South Korea, which managed to make the top ten list in three consecutive decades—1970s, 1980s, and 1990s—before slowing down in the 2000s was also very impressive.

More broadly, table 3 demonstrates the dominating growth performance of Asian economies. Since the 1970s, almost half of the top ten growth performers come from Asia, a truly remarkable achievement. On

TABLE 3
COUNTRY RANKINGS FOR EACH DECADE BY PER CAPITA GDP GROWTH (%)

Decade	Rank	Country	Decade avg. GDP (per capita) growth (%)	Rank	Country	Decade avg. GDP (per capita) growth (%)
1960s	1	Japan	8.98	93	China	-0.32
	2	Mauritania	8.16	94	Rwanda	-0.74
	3	Greece	7.75	95	Algeria	-0.79
	4	Romania	7.73	96	Mauritius	-0.96
	5	Morocco	7.68	97	Haiti	-1.51
	6	Hong Kong	7.48	98	Guinea	-1.58
	7	Spain	6.92	99	Senegal	-1.76
	8	Iran	6.51	100	Nigeria	-2.13
	9	Cyprus	6.47	101	Bangladesh	-2.14
	10	Portugal	5.92	102	Mali	-2.25
1970s	1	Botswana	10.85	111	Madagascar	-1.44
	2	Romania	8.99	112	Central Africa	-1.69
	3	Singapore	7.76	113	Liberia	-1.85
	4	Iraq	7.75	114	Congo, Dem. Republic	-1.94
	5	Korea	7.19	115	Chad	-3.32
	6	Malaysia	7.1	116	Uganda	-3.84
	7	Swaziland	6.82	117	Zambia	-3.88
	8	Hong Kong	6.71	118	Nicaragua	-4.25
	9	Bulgaria	6.51	119	Lebanon	-5.2
	10	Indonesia	6.34	120	Cambodia	-6.52
1980s	1	China	7.34	116	Togo	-3.27
	2	Botswana	6.93	117	Venezuela	-3.33
	3	Korea	6.54	118	Iraq	-4.17
	4	Egypt	5.61	119	Nigeria	-4.19
	5	Hong Kong	5.25	120	Libya	-4.38
	6	Thailand	5.11	121	Bahrain	-4.6
	7	Cyprus	4.61	122	Niger	-4.72
	8	Singapore	4.55	123	Iran	-5.11
	9	Kuwait	4.22	124	Lebanon	-5.13
	10	Lao	4.08	125	Trinidad and Tobago	-5.16
1990s	1	China	8.64	138	Azerbaijan	-4.63
	2	Lebanon	7.22	139	Russia	-5.27
	3	Ireland	6.02	140	Moldova	-5.93
	4	Armenia	5.96	141	Ukraine	-6.72
	5	Vietnam	5.39	142	Liberia	-7.11
	6	Eritrea	5.31	143	Sierra Leone	-7.23
	7	Chile	5.11	144	Serbia	-7.93
	8	Korea, Republic	4.94	145	Afghanistan	-9.1
	9	Malaysia	4.88	146	Tajikistan	-9.9
	10	Singapore	4.31	147	Congo, Dem. Republic	-10.85
2000s	1	Azerbaijan	13.19	139	Guinea	-0.35
	2	Kazakhstan	9.2	140	Congo, Republic	-0.45
	3	China	9.13	141	Madagascar	-1.09
	4	Armenia	8.26	142	Togo	-1.14
	5	Trinidad and Tobago	8.25	143	Gabon	-1.48
	6	Afghanistan	8.08	144	Central Africa	-1.53
	7	Belarus	7.81	145	Timor-Leste	-1.73
	8	Angola	7.65	146	Cote d'Ivoire	-1.82
	9	Albania	7.22	147	Zimbabwe	-3.4
	10	Lao	6.59	148	Eritrea	-4.62

Source: Penn World Tables version 7.1

the other hand, some of the poorest countries in the world, many of them fragile states, continued to record negative growth rates decade after decade. The encouraging news for developing economies emerges in the 2000s, when unexpectedly some LICs make the top ten list for the first time in decades. Unfortunately, consistent with the very poor growth performance among fragile states discussed previously, table 3 also shows that the bottom ten list is mainly composed of these countries.

2.7 Convergence of LICs to Middle-Income Status

How many years would it take LICs to reach MICs' income levels? To answer this question, we consider a scenario in which we assume that the average annual growth rate for each country in the past decade will prevail for the indefinite future. Then we calculate the number of years that would be required for each LIC to reach the middle-income threshold of a little over \$3,000. Table 4 reports these calculations and lists countries in ascending order of years to middle-income status. According to this simple approach, which is arguably quite optimistic in nature, Vietnam and Laos have almost reached middle income status while Moldova, Sudan, and Cambodia are within reach in less than ten years. Countries in East Africa that exhibit strong growth rates nowadays, such as Rwanda, Tanzania, and Uganda, would need about three decades to reach middle-income status, whereas Congo, Benin, and Sierra Leone are far behind, being a century or more away based on recent growth performance. This exercise serves as a reminder that huge cross-country income disparities are not likely to be easily mitigated, even under favorable assumptions.

2.8 Episodic Growth

An important fact in the literature of economic development is that growth in many countries, and particularly in LICs, is

TABLE 4
NUMBER OF YEARS REQUIRED FOR SELECTED LOW-INCOME COUNTRIES TO ACHIEVE MIDDLE-INCOME STATUS

Country	Year
Vietnam	2.5
Lao	3.7
Moldova	6.5
Sudan	7.6
Cambodia	9.5
Ghana	13.1
Kyrgyzstan	14.4
Papua New Guinea	14.8
Tajikistan	18.1
Nigeria	20.0
Nicaragua	22.0
Bangladesh	24.9
Mauritania	27.2
Liberia	28.7
Rwanda	33.9
Uganda	36.5
Nepal	43.0
Senegal	46.2
Cameroon	47.7
Mali	50.9
Burkina Faso	51.7
Malawi	86.8
Gambia	90.0
Congo	95.6
Benin	118.0
Sierra Leone	120.1
Haiti	149.5
Guinea	228.9
Niger	734.3

Note: Future growth projections are based on the average per capita GDP growth over the period 1995–2010. Countries with negative growth rates 1995–2010 are not reported.

Source: Penn World Tables version 7.1.

highly episodic and characterized by periods of accelerated growth followed by sharp decelerations, often leading to disasters. Figure 7a illustrates this claim by plotting the growth rate of each developing and advanced country in our sample for each of the 1970s, 1980s, 1990s, and 2000s against that in the previous decade. The

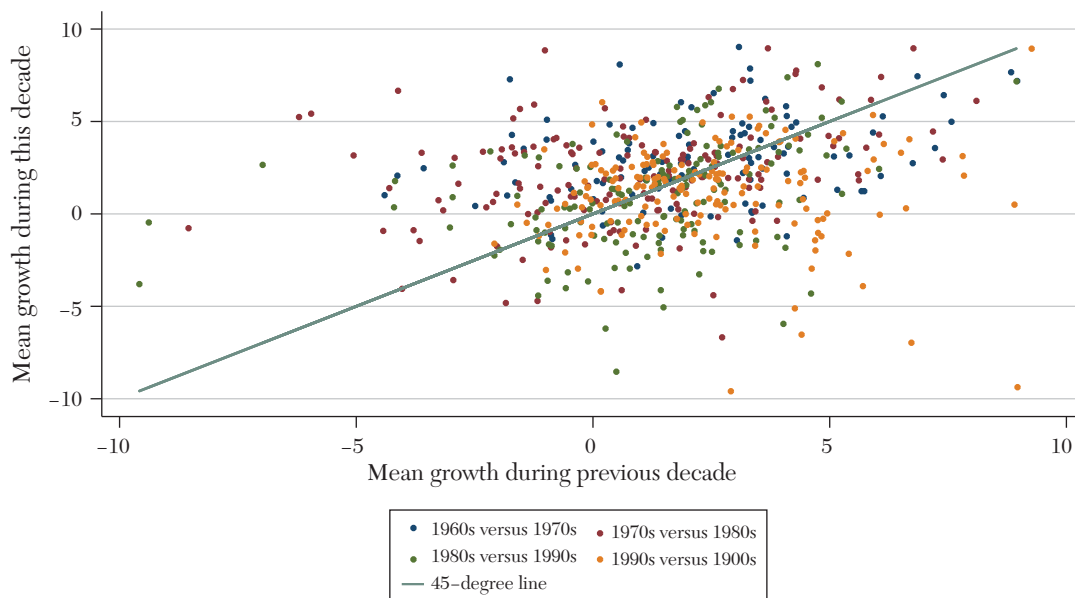


Figure 7a. Correlations of per Capita GDP Growth in Consecutive Decades (All Countries)

Notes: Income definition based on PPP converted GDP per capita (chain series), at 2005 constant prices. First decade growth rate showed on x axis.

Source: Penn World Tables 7.1.

wide dispersion of these points around the forty-five-degree line reflects the unpredictability of growth from one decade to another. Figure 7b demonstrates that the instability of growth across decades is even larger in LICs. Over the past few decades, the typical growth rate for a LIC in one decade has generally been a poor predictor of its growth rate during the next decade, while many policies and country characteristics have been more stable (Easterly et al. 1993).^{11,12}

¹¹The correlations coefficient in LIC growth rates for the 1960s versus 1970s is 0.011, for the 1970s versus 1980s is -0.118, for the 1980s versus 1990s is 0.025, and for the 1990s versus 2000s is -0.212.

¹²While this type of growth nonlinearity is related to the existing theoretical literature on poverty traps, it is important to recognize that it is also quite different in that

The empirical literature on growth spells took off after Pritchett (2000) demonstrated that the growth process, especially in developing economies, tends to be episodic. Hausmann, Pritchett, and Rodrik (2005) proposed a heuristic approach for identifying growth breaks while more recent research has focused on improving the methodology for obtaining growth breaks. For example, Jerzmanowski (2006) estimates Markov-switching regressions to characterize four distinct growth regimes and

countries need not fall into persistent underdevelopment once they experience a growth down break. Rather, the experience has been that countries tend to grow in spells and even for the very poor countries growth is not constantly low but rather start-and-stop.

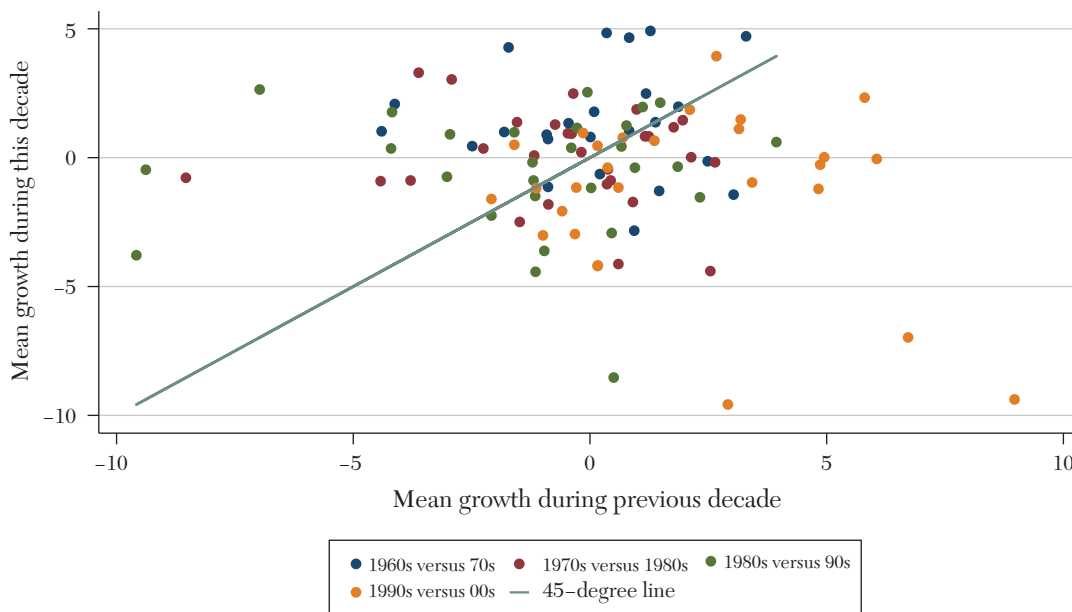


Figure 7b. Correlations of per Capita GDP Growth in Consecutive Decades (LICs)

Notes: Income definition based on PPP converted GDP per capita (chain series), at 2005 constant prices. First decade growth rate shown on x axis. Coefficient of correlation (Pearson): 1960s vs. 1970s (0.011); 1970s vs. 1980s (−0.118); 1980s vs. 1990s (0.025); 1990s vs. 2000s (−0.212).

Source: Penn World Tables 7.1.

transitions between them. Jones and Olken (2008) use the structural break econometric technique of Bai and Perron (1998, 2003), which locates and tests for multiple structural breaks within a time series to identify dates of turning points. Berg, Ostry, and Zettelmeyer (2012) turn attention to the challenge of sustaining growth accelerations by searching for determinants of the duration of growth spells.

Table 5 presents spells of growth accelerations and deceleration across decades using the methodology developed in Berg, Ostry, and Zettelmeyer (2012) and data from PWT 7.1. This table confirms that the growth experience in many countries is not smooth, but rather erratic, with the table

featuring almost as many accelerations as decelerations. Both kinds of growth spell have been observed in all five decades in our sample, but the 1970s are dominated by decelerations and the 1990s by accelerations. It is also evident that growth is most episodic in developing economies (LICs and MICs), with there being only a few examples of growth spells of either kind in advanced economies. In terms of geographical dispersion, countries in sub-Saharan Africa are the most prone to episodic growth. This evidence suggests that whether convergence or divergence characterizes the long-run behavior of growth, the process is highly fragmented in developing and low-income economies.

TABLE 5
GROWTH BREAKS BY DECADE

Decade	Accelerations		Decelerations	
1960	Guatemala	1962	Hong Kong	1965
	Indonesia	1967	Morocco	1963
	Korea	1962	Togo	1969
	Pakistan	1960	Zambia	1968
	Portugal	1967		
1970	Bangladesh	1975	Austria	1979
	Cameroon	1973	Belgium	1974
	China	1977	Botswana	1973
	Ecuador	1970	Cameroon	1978
	Egypt	1976	Cote d'Ivoire	1978
	Gabon	1971	Ecuador	1978
	Mauritius	1971	El Salvador	1978
	Paraguay	1973	France	1974
	Vietnam	1975	Gabon	1976
			Greece	1973
			Hungary	1978
			Indonesia	1973
			Iran	1976
			Israel	1973
			Italy	1974
			Jamaica	1972
			Japan	1970
			Mauritius	1977
			Poland	1979
			Portugal	1973
			Romania	1978
			Spain	1974
			Swaziland	1975
			Switzerland	1970
			Tanzania	1971
1980	Cambodia	1982	Albania	1988
	El Salvador	1984	Brazil	1980
	Guatemala	1987	Bulgaria	1988
	Iran	1981	Cameroon	1984
	Jamaica	1980	Egypt	1984
	Tanzania	1985	Guatemala	1980
	Uganda	1988	Mexico	1981
			Paraguay	1980
			Romania	1987
			South Africa	1980
			Trinidad and Tobago	1980

(Continued)

TABLE 5
GROWTH BREAKS BY DECADE (*Continued*)

Decade	Accelerations		Declerations	
1990	Albania	1992	Eritrea	1998
	Angola	1993	Japan	1991
	Bangladesh	1996	Korea	1996
	Bulgaria	1997	Kuwait	1993
	Cameroon	1994	Norway	1998
	Czech Republic	1992	Sierra Leone	1994
	Ireland	1993	Slovak Republic	1990
	Kazakhstan	1998	Tajikistan	1999
	Kyrgyzstan	1995	Tanzania	1990
	Liberia	1996	Thailand	1995
	Moldova	1994		
	Mozambique	1995		
	Poland	1991		
	Romania	1992		
	Serbia	1993		
	Sierra Leone	1999		
	Slovak Republic	1992		
	South Africa	1993		
	Tajikistan	1997		
	Tanzania	1997		
	Trinidad and Tobago	1993		
	Turkmenistan	1995		
	Uzbekistan	1994		
	Zambia	1994		
2000	Ethiopia	2003	Ireland	2003
	India	2002	Kazakhstan	2006
	Kazakhstan	2000	Liberia	2000
	Laos	2004		
	Mongolia	2002		
	Uzbekistan	2003		

Based on these very basic facts, broad-based international convergence is hard to witness, but there are countries that have enjoyed very successful convergence experiences. Those typically come from the middle stages of economic development, and mostly from South and East Asia, with notable examples being China and South Korea—China managed a leap from negative growth in the 1960s to become the highest growing county in the world in the course of the next few decades, while South Korea was propelled from low-income status

to high-income status within the short span of fifty years. However, the majority of the poorer countries, mostly in Africa but elsewhere too, show no movement in closing the gap that has been increasing as more advanced economies grow at a faster pace (figure 3).

3. Theoretical Considerations

The convergence hypothesis is often taken to be the proposition that per capita

incomes in different countries are somehow getting closer to each other, but it is more precisely understood as the proposition that the long-run cross-country distribution of per capita income is independent of initial conditions. This was originally conceptualized in the basic Solow–Swan neoclassical growth model where, outside of the steady state, economic growth is driven by the accumulation of physical capital. As the marginal product of capital is diminishing, this accumulation should come to a stop so that, as a result, rich countries’ growth will slow allowing poorer countries to catch up. Much of modern growth economics introduces other sources of heterogeneity, notably technological progress (both innovation and adoption) and human capital accumulation as well as differences in geography, institutions, and policy, which may slow down or speed up the process of convergence.

In this section, we briefly review the theoretical underpinnings of convergence by providing a sketch of how dynamics should be thought about under the basic Solow model and other more modern theories. The section is then rounded off with “economically interesting” notions of convergence that fall into two logically distinct convergence concepts that are predominant in the literature: one suggesting that disparities between countries simply disappear with time; the other advocating that convergence is inherently dependent on a given country’s limiting behavior from certain initial conditions. This naturally leads to the question of absolute versus conditional dissipation of initial disparities.

The original theoretical foundation of the convergence hypothesis is the unique stable steady state in the standard one-sector neoclassical growth model. This property follows from the standard conditions on the production function of a strictly diminishing marginal product of capital that takes all values between infinity and zero as the capital

stock rises from zero to infinity. Ignoring population growth and technological progress, in the Solow–Swan version of the neoclassical model with an exogenous saving rate of s , the growth rate of capital per worker, k , is given by $g_k = sf(k)/k - \delta$ where $f(k)$ is the (intensive form of the) production function and δ is the depreciation rate. The assumptions on $f(k)$ ensure that $f(k)/k$ declines monotonically from infinity to zero as k rises so that $g_k > 0$ for small positive k pushing k toward the single value of k , independent of the initial quantity of capital per worker, where $sf(k)/k = \delta$. This ensures the existence of a unique, stable steady state that is eventually reached for any initial $k > 0$ and it is in this sense that the long-run outcome of the economy is independent of its initial conditions.

To move beyond the basic Solow–Swan model, we can draw on growth and development accounting calculations showing that growth is driven by the accumulation of both human and physical capital as well as improvements in technology or total factor productivity. To fix ideas, suppose that the aggregate production function can be written as $Y = AF(K, hL)$ where Y is GDP, A is the level of technology, K is the stock of physical capital, L is the labor force, h is human capital per worker, and F is a production function, which we assume to exhibit constant returns to scale in its two arguments. Using that assumption, we can write per capita output as $y = AF(k, h)$ where $y = Y/L$ and $k = K/L$. As in the basic Solow–Swan model, the assumption of diminishing returns to the accumulable factors of production k and h yields convergence to a steady state and the addition of the assumption of a common level of technology A implies that the steady state is common across countries.¹³

¹³See, for example, the exposition in Durlauf and Johnson (1995).

The assumption of diminishing returns to human capital is justified by the observation that higher education is bounded from above and by studies in labor economics finding that human capital's contribution to income is diminishing (see, e.g., Becker 1994).¹⁴ Nonetheless, as a voluminous literature has shown, this is not clear at all with respect to productivity improvements stemming from technological innovation and adoption (see, e.g., Schumpeter 1950, Aghion and Howitt 1992, Zeira 1998). The assumption of a common level of technology is not innocuous but was implicit in much of the early convergence literature.¹⁵ As we discuss below, more recent work has emphasized the inability of some countries to reach the technological frontier as an obstacle to convergence.

To go beyond the exogenous technological progress of the basic Solow–Swan model, we follow a very simple formulation proposed in Aghion (2004). In this model, a country's productivity A grows over time according to $\dot{A} = \lambda\varphi(A^{\max} - A)$, where $A^{\max}$ is the worldwide technological frontier and $\lambda\varphi$ is the rate of creative destruction (λ measures the productivity of research and development (R&D); φ is the R&D intensity, measured as the productivity-adjusted quantity of final output devoted to R&D). Defining $a = A/A^{\max}$ and letting g denote the growth rate of $A^{\max}$ yields $\dot{a} = \lambda\varphi(1 - a) - ag$ as the differential equation describing knowledge transfers generate convergence to the global growth rate. Similar to the law of diminishing returns in the neoclassical growth model, as

the gap between the country's average productivity and the worldwide leading-edge, $1 - a$, narrows, innovations will raise productivity at a decreasing rate.

Embedded in a version of the Solow–Swan model with technological progress, this setup yields $g_k = sf(k)/k - [\delta + \lambda\varphi(a^{-1} - 1)]$ as the rate of the growth rate of capital per intensive worker, $k = K/AL$. While this equation seems similar to that above derived from the standard Solow–Swan model, there are two drivers of convergence behavior here. As before, there is the declining marginal product of capital but, in addition, there is the adoption of technologies that propel growth by pushing the level of technology in a country toward the worldwide technological frontier. Perhaps more importantly, this model also admits the possibility of convergence club behavior. In countries where there are sufficiently high barriers to technological innovation or adoption due to, for example, corrupt institutions, poorly defined intellectual property rights, low supply of skilled workers, and credit constraints (see Aghion et al. 2016, Howitt and Mayer-Foulkes 2005), φ will be zero. Such countries will be left behind while those countries with positive φ move toward the technological frontier, thus creating two groups of countries with group membership dependent on initial conditions.

Despite its fundamental insights, the elementary Solow–Swan model suffers at least one potentially important drawback in the convergence context in that it is a closed-economy model while most countries, even some of the poorest and the smallest, have, at least to some extent, borders open to trade and capital flows. Some of the recent literature has introduced open economy growth models that consider trade and capital flows to be elemental drivers of globalization and growth. Examples of such models that consider the issue of convergence include Barro, Mankiw,

¹⁴As a voluminous literature has shown, it is far from clear that this assumption is warranted with respect to productivity improvements stemming from technological innovation and adoption (see, e.g., Schumpeter 1950, Aghion and Howitt 1992, Zeira 1998).

¹⁵Mankiw (1995, p. 301), for example, argues that for “understanding international experience, the best assumption may be that all countries have access to the same pool of knowledge, but differ by the degree to which they take advantage of this knowledge by investing in physical and human capital.” Romer (1993) offers a contrasting view.

and Sala-i-Martin (1995); Ventura (1997); Cuñat and Maffezzoli (2004); and Oxborrow and Turnovsky (2017). Of course, in an open-economy growth model with perfect international capital mobility convergence would happen instantly because incipient cross-country differences in rates of return on capital would be immediately eliminated. To explain why we do not observe equalization of cross-country return on capital and therefore immediate convergence, we need to consider possible frictions in international capital markets that slow convergence or eliminate it all together.

To exposit these ideas, we use an elementary open economy model, from Obstfeld and Rogoff's (1996) textbook, which allows for international borrowing and lending under credit market imperfections. In this overlapping generations model, a small open economy faces a fixed world interest rate r . Individuals in this economy live two periods, working only in the first period and earning wage w_t . Individuals can borrow in the world capital market an amount b_t (up to a fraction $\eta > 0$ of their earnings). Savings is based on individual choice rather than being a constant fraction of income as in the Solow–Swan model.

As in the standard neoclassical model, the equilibrium domestic interest rate r^d is equal to net marginal return to domestic investment, $r^d = f'(k) - \delta$. This rate can exceed the world rate r if the international borrowing constraint is binding. An individual at the first period of life maximizes the standard logarithmic utility function, $U_t = \log(c_t) + \theta \log(c_{t+1})$, subject to the constraints $k_{t+1} + b_{t+1} = w_t - c_t$ and $c_{t+1} = (1 + r_{t+1}^d)k_{t+1} + (1 + r)b_{t+1}$ where c_t is the individual's consumption in period t and b_{t+1} is an individual's assets abroad that are subject to the constraint $b_{t+1} \geq -\eta w_t$. For simplicity of exposition, we omit the details of this standard maximization problem and proceed as in Obstfeld and Rogoff (1996)

with considering the three possible cases that are implied by the model for a country that opens itself to world capital markets after having reached its autarkic steady state.

If $r^d < r$ initially, then the small open economy becomes a net creditor in the international capital market and r^d converges to r in one period implying absolute and rapid convergence in the steady state. However, if $r_{t+1}^d > r$, the borrowing constraint will bind so that $b_{t+1} = -\eta w_t$, which yields

$$(2) \quad k_{t+1} = w_t - c_t - b_{t+1} \\ = \left[\frac{\theta(1 + \eta)}{1 + \theta} + \frac{(1 + r)\eta}{(1 + \theta)(1 + r_{t+1}^D)} \right] w_t.$$

It is interesting to note that setting $\eta = 0$ the model collapses to one resembling a closed economy Solow growth model, albeit with log utility maximizing agents (and noticing that w_t represents savings out of output). Equation (2) implies that greater capital inflows from international markets would speed up convergence because easing the borrowing constraint would lower r^d , thereby increasing the rate of capital accumulation. Letting k_{ss}^d denote steady-state capital stock under constrained borrowing and k_{ss}^u that in the absence of a borrowing constraint, one can show that $k_{ss}^d > k_{ss}^u$ is the same as $k_{ss}^u < (\varphi w_{ss}/(1 + \varphi)) + \eta w_{ss}$ where w_{ss} is the wage of the agent in the first period of life in the unconstrained steady state. This condition offers an intuitive explanation of convergence dynamics in this model, as it states that if the individuals' saving (w_{ss}) plus maximum amount possible from international capital markets are sufficient to finance k_{ss}^u , then the economy will achieve convergence to the unconstrained steady state k_{ss}^u . If, however, this inequality is reversed the economy will not ever converge to steady state k_{ss}^u simply because the wage and maximum possible foreign borrowing would not be sufficient to finance k_{ss}^u .

In sum, this model demonstrates in a very tenable and intuitive way how market imperfections can deliver convergence dynamics that are compatible with the reality of more integrated global economy and consistent with evidence from the various empirical approaches that we discuss below.¹⁶

These and other theoretical growth models provide a variety of sources of heterogeneity in long-run outcomes. The empirical challenge is to determine if that heterogeneity represents the long-run effect of initial conditions and so is at odds with the convergence hypothesis, or if it merely represents cross-country microeconomic variation. An example of latter case would be something that can be conceived of as parameter variation in the Solow–Swan model, such as different exogenous saving rates, the removal of which would permit a common long-run outcome in the absence of differences in initial conditions. In the former case, the obstacle to convergence between two countries would be membership of different basins of attraction of the process describing the evolution of per capita income so that the initial conditions defining that membership have long-run effects. We now turn to an outline of the empirical convergence concepts that have been employed in response to this challenge.

4. Convergence Concepts

The most elementary convergence tests are the so-called β -convergence tests. Often these tests use a log-linearized version of the neoclassical growth model to motivate estimation of equations of the form

$$(3) \log(y_t/y_{t-\tau}) = \alpha + \beta \log(y_{t-\tau}) + u,$$

¹⁶In developing a theory of gradual adjustment of capital in an open economy, an alternative to imperfect capital markets model presented above is the model of convex adjustment costs to investment and Tobin's q (see, e.g., Battisti, di Vaio, and Zeira 2016).

where $\tau > 0$, α and β are parameters, and u is an error term. A test of the hypothesis $\beta = 0$ against the alternative $\beta < 0$ is then construed as a test for convergence, as the neoclassical model implies that a country will grow more quickly the further it is from the steady state, which is implied by $\beta < 0$. This approach to testing the convergence hypothesis has been widely implemented as a cross-section test for a group of countries as well as a panel test for a group of countries over time. Durlauf, Johnson, and Temple (2005) contains an extensive survey of this literature and discusses at length its econometric and conceptual pitfalls.

As a test of convergence as the irrelevance of initial conditions, such β -convergence tests require that initial income be a sufficient statistic for a country's initial conditions. To the extent that initial conditions are also reflected in cross-country variation in α , these tests can have low power against non-convergent alternatives, a point originally developed in Bernard and Durlauf (1996). Several studies, among them Durlauf and Johnson (1995), Tan (2010), and Fiaschi, Lavezzi, and Parenti (2019) have found that variables other than initial income (literacy, institutional quality, ethnic fractionalization, life expectancy, share of Catholics) are capable of defining groups of countries in which exhibit similar within-group long-run behavior but different across-group long-run behavior. Such results imply that initial income does not contain all of the information needed to determine the long-run distribution of a country's per capita output.

While the neoclassical model is the motivation for tests of the convergence hypothesis in early work such as Barro (1991), Barro and Sala-i-Martin (1992), and Mankiw, Romer, and Weil (1992), still earlier work such as Abramovitz (1986) and Baumol (1986) was motivated by considerations of technology transfers and capital flows from "leader" to

“follower” countries.¹⁷ Under this view, countries that are further behind the leader are potentially able to make a larger leap forward and grow relatively faster than the leader as they catch up. This is a potentially important driver of convergence, and Sachs and Warner (1995) cite a lack of openness as major obstacle to development and hence, convergence. Dowrick and DeLong (2003) argue that while periods of increased globalization such as those prior to World War I and after World War II tended to foster convergence in that the “convergence club” tended to grow in these periods, the effects were far from universally felt. They note that many countries were unable to join the club or, if they did, were unable to maintain foothold in it. They reexamine Sachs and Warner’s estimates of the effect of openness on growth and conclude that while openness does promote growth, the benefits of doing so seem to have declined since 1980.¹⁸

Following Barro (1991) and Mankiw, Romer, and Weil (1992), equation (3) is sometimes augmented with a term such as $\lambda'x$, where λ is a vector of parameters and x is a vector of conditioning variables that determine the steady-state value of output per capita—variables such as rates of physical and human capital accumulation and population growth as well as a wide variety of

others.¹⁹ In this case, countries are assumed to have different steady states only because of the microeconomic variation controlled for by the inclusion of x and a negative estimated value of β is taken as evidence that each is converging to its particular steady state. Such tests are called tests of “conditional convergence” to distinguish them from tests of “absolute convergence” based on equation (3).²⁰

The implication of this finding of conditional convergence is that a poor country can be made to converge to prosperity simply by adopting the value of x of a rich country. Dowrick and DeLong (2003, p. 204) describe the presumption that a poor country could do this as the “joker in the deck,” arguing that “a moment’s thought will convince anyone that many of the right-hand-side variables used by Barro (1997) could never be brought to the mean values found in the industrial core of the world economy in any country that has not already attained the productivity level and socioeconomic structure found in the industrial core.” There is another critique of the conditional convergence concept that is a bit more technical in character, yet equally

¹⁹The inclusion of these variables raises important concerns about endogeneity as discussed by Cho (1996), Temple (1999), Easterly (2005), Durlauf, Johnson, and Temple (2005), Rodrik (2012), and Lenkoski, Eicher, and Raftery (2014), among others.

¹⁷While not concerned primarily with the convergence hypothesis per se, another early study, Kormendi and Meguire (1985), includes initial income levels in their growth regressions motivated by the neoclassical model and note the catching-up implications of the negative estimated coefficient. As they point out, Barro (1984, pp. 288–94) studies graphically the catching-up behavior of a small group of industrialized countries in the postwar period. See also Marris (1982).

¹⁸Alesina, Spolaore, and Wacziarg (2005) find that openness and country size are substitutes in promoting growth. Di Vaio and Enflo (2011) discuss the literature on theoretical and empirical motivations for range of possible effects of globalization on the cross-country distribution of income.

²⁰While Mankiw, Romer, and Weil (1992) derive a set of x variables from a version of the Solow growth model, economic theory is largely silent with respect to the set of variables to be included. As a result, empirical studies often abuse the resulting flexibility for selecting among the potential candidates. So much so that Durlauf and Quah (1999) report that over ninety different variables have been used despite the fact that no more than 120 country observations were available for the regression analysis using early versions of Penn World Data (version 4.0). This was part of the motivation for the literature aimed at eliminating model uncertainty (see, e.g., Levine and Renelt 1992; Raftery 1995; Doppelhofer, Miller, and Sala-i-Martin 2000; Fernández, Ley, and Steel 2001; Brock and Durlauf 2001; Ley and Steel 2009b; Eicher, Papageorgiou, and Raftery 2011; Leamer 2016a, b). One of the key findings of this literature is that the initial per capita GDP level is the most effective of all variables tried in explaining growth.

powerful: if indeed per capita GDP converges to different steady states, then the income distribution itself should converge to a limit distribution, which is not consistent with the evidence presented in the empirical section above. For these reasons, we mostly consider the absolute version of the convergence in this article.

While divergence is an obvious alternative to convergence, another economically interesting possibility is club convergence, in which groups of countries with similar initial conditions exhibit similar long-run outcomes so that, for example, the cross-country distribution of per capita income can have two or more peaks, as found by Quah (1993a). This can reflect a law of motion for the evolution of output per capita that has more than one stable steady state as can occur in a model with a feedback loop from income to some other state variable. In such cases, club convergence can occur as economies converge to the steady state associated with the basin of attraction in which they begin. A country's long-run per capita income will then depend on its initial conditions as measured by the state variables determining its basin of attraction with countries having similar initial conditions having similar long-run per capita income, and so forming a convergence club.

Bernard and Durlauf (1996) and Durlauf and Johnson (1995) have argued that the β -convergence tests have low power against this type of alternative, with the latter paper finding that the club convergence hypothesis is consistent with the Mankiw et al. (1992) data. Following Quah (1993a) and Durlauf and Johnson (1995), a large group of authors have examined the convergence hypothesis using methods that have club convergence as the alternative hypothesis. Some of this research follows Quah and studies the dynamics of the entire cross-country distribution of per cap-

ita income, while other researchers have used a variety of clustering approaches to divide their samples into groups of countries that represent putative convergence clubs because of similarities in initial conditions.

The absence of a role for initial conditions in long-run outcomes implies that contemporary differences in per capita incomes are transitory, suggesting that the dispersion of per capita incomes across economies should fall if convergence is occurring. Barro and Sala-i-Martin (1992) refer to this notion as σ -convergence, which is said to occur between t and $t + \tau$ if $\sigma_t^2 > \sigma_{t+\tau}^2$, where σ_t^2 is the cross-country variance of y_t . While β -convergence is not sufficient for σ -convergence, as shocks can cause σ_t^2 to be constant or increase over time, even if β -convergence is occurring, Young, Higgins, and Levy (2008) show that β -convergence is necessary for σ -convergence. Friedman (1992), Quah (1993b), and Hart (1995) caution against committing Galton's fallacy, as β -convergence may be observed even if there is a constant, or even increasing, variance between two points in time. Accordingly, Friedman (1992), Hart (1995), Lichtenberg (1994), and Carree and Klomp (1997) emphasize that tests for convergence should investigate whether the variance indeed decreases between two points in time, i.e., whether there is σ -convergence (Barro and Sala-i-Martin 1995).

A different approach is taken by Bernard and Durlauf (1995, 1996) who offer definitions of convergence based on the time series behavior of output. They represent the idea that initial conditions have no implications for a country's per capita income level in the long run by saying that two countries converge if the current long-run forecasts of their log per capita income levels are equal.²¹

²¹ This equality holds as the absence of a role for initial conditions in determining the long-run distributions

That is, if

$$(4) \quad \lim_{T \rightarrow \infty} E(y_{i,t+T} - y_{j,t+T} | F_t) = 0,$$

where $y_{i,t}$ denotes the log of per capita income in country i at time t and F_τ denotes the history of $y_{i,t}$ and $y_{j,t}$ up to time τ . This definition implies that the deviation between the two countries is expected to decrease: $E(y_{i,t+T} - y_{j,t+T} | F_t) < y_{i,t} - y_{j,t}$ for some T when $y_{i,t} > y_{j,t}$, so that convergence can be thought of as “catching up” also in the time series context. As definition (4) implies the absence of stochastic or deterministic trends in the cross-country difference of log per capita income levels, convergence has often been tested in the time series context by testing the stationarity of that difference. For countries with log of per capita income obeying integrated processes, this test can be implemented as a test of the cointegration of country pairs of $y_{i,t}$ and $y_{j,t}$ with cointegrating vector $[1, -1]$. Applied to a group of such countries, this notion of convergence implies that the log per capita income levels are generated by a process with a single stochastic trend, although that alone is not sufficient to imply convergence in the sense of definition (4) above. Bernard and Durlauf (1995) conclude that, while the number of common trends among fifteen industrialized countries over the period 1900–1987 is small, it exceeds one and so they can not reject the null hypothesis of no convergence despite De Long’s (1988) acknowledged observation that use of this sample ought to bias the results toward a finding of convergence. As we describe later on, there have arisen variations of this time-series approach to convergence.

of per capita output in two countries implies that the two countries will have identical limiting distributions of per capita output. See Durlauf, Johnson, and Temple (2005) for details.

5. Empirical Evidence

This section visits the recent evidence on the convergence hypothesis. While we focus on the research of the last decade or so that has examined the hypothesis using cross-country data rather than data on regions as economic units, earlier work is briefly mentioned in places to provide appropriate context.²²

5.1 Linear Models

The early contributions to the convergence literature estimate versions of equation (3) above and test for a negative β , or equivalently, a negative correlation between initial per capita income and its subsequent growth rate. Baumol (1986), Barro (1991), Dowrick (1992), and others fail to find a negative correlation when the sample is a broad group of countries, implying a rejection of the absolute convergence hypothesis.

A striking result by Rodrik (2013) regarding unconditional convergence suggests that unlike economies as a whole, manufacturing industries exhibit strong unconditional convergence in labor productivity. The result holds at various levels of disaggregation for a large sample covering more than one hundred countries over recent decades. Rodrik’s interpretation of this result is that sustaining growth requires active policies that promote economic diversification and structural change from low-productivity activities to mostly tradable higher-productivity activities. The challenge is then to identify these automatic-convergence industries in each country and to expand domestic employment

²²Earlier discussions and examinations of the convergence hypothesis can be found in review articles such as Temple (1999), Islam (2003), and Abreu et al. (2005), the *Handbook of Economic Growth* volumes I and II by Aghion and Durlauf (2005, 2014, respectively), and in various textbooks including Grossman and Helpman (1991), Barro and Sala-i-Martin (1995), Jones (1998), Aghion and Howitt (2009), and Acemoglu (2009).

around these high-productivity industries. Of course, as Rodrik readily admits in several of his recent writings on growth and convergence (2011, 2013, 2014, 2015), the problem is that mobilizing the productive sectors most often requires hard economic choices and even harder political decisions that rarely amount to what is necessary to get structural change going in many developing economies and especially in LICs.

Abramovitz (1986) emphasized that “social capabilities” including the ability to absorb existing technologies and attract capital are prerequisites and must be in place in an economy before catch-up growth can take place. Contributions such as Barro (1991), Barro and Sala-i-Martin (1992), Dowrick (1992), and Mankiw, Romer, and Weil (1992) show that the addition of conditioning variables such as rates of capital accumulation, population growth rates, and policy variables renders a statistically significant negative partial correlation between initial per capita income and its subsequent growth rate, evidence consistent with the conditional convergence hypothesis. Notably, Mankiw et al. (1992) was the first contribution to derive the set of conditioning variables from an explicit growth model. Indeed, the foundational papers by Barro and Sala-i-Martin (1992) and Mankiw, Romer, and Weil (1992) initiated a huge literature attempting to empirically test the extent of conditional β -convergence in various contexts.

A likely influential culprit impeding nations from converging is openness to international markets. According to Sachs and Warner (1995),²³ convergence is not occurring everywhere because of the closed economic policy of some developing countries. According to these authors “...open economies tend to converge, but closed economies do not. The lack of convergence in recent decades results

from the fact that the poorer countries have been closed to the world.” The original results obtained by these studies were generally affirmed by studies using panel data such as Islam (1995) and Caselli, Esquivel, and Lefort (1996). Subsequent contributions also introduced various adjustments to the original estimation strategy including a spatial dimension (see, e.g., Baumont, Ertur, and Le Gallo 2003 or Dall’erba and Le Gallo 2006) to mitigate concerns that the omission of space from the analysis of the β -convergence process could produce biased results.

As pointed out by Sala-i-Martin (1996), one of the striking results obtained in these studies is that the speed of convergence with which economies converge to their steady state is roughly 2 percent a year. To this day, the 2 percent convergence rate continues to make headlines. For example, in recent work Barro (2015) shows that in a panel of countries since 1960 the estimated annual convergence rate for GDP is 1.7 percent, conditional on various explanatory variables. With data starting in 1870, he estimates the convergence rate at 2.6 percent; therefore, combining the two estimates Barro calculates conditional convergence close to what he calls the “iron-law” rate of around 2 percent (see also Lee 2017, who considers convergence in the context of Chinese growth experience).

Similarly, the restriction of the sample to a group of sufficient similar countries such as the industrialized countries (Abramovitz 1986, Baumol 1986, Dowrick and Nguyen 1989) or the individual states of the United States (Barro and Sala-i-Martin 1991, 1992) also yields a negative correlation between initial per capita income and its subsequent growth rate. Importantly, DeLong (1988) points out that Baumol’s finding of convergence in a group of ex post successful countries may reflect sample selection issues as unsuccessful countries that have thus not converged were excluded from the group

²³A Google Scholar search reports 5,628 citations of this paper as of October 2016.

of countries studied in Madison (1982), the source of Baumol's data.

Bernard and Durlauf (1996) question the power of the β -convergence test based on equation (3) and argue that rejection of the null hypothesis that $\beta = 0$ in favor of the alternative that $\beta < 0$ is likely, even if the data are generated by a model with multiple steady states. This means that, even if the data are generated by a model in which countries follow locally linear but globally nonlinear laws of motion, the standard β -convergence test may still lead a researcher to conclude that convergence to a single steady state is occurring. In other words, the standard test has relatively low power against the alternative of multiple steady states.

To improve the power of the β -convergence test against this alternative, Durlauf and Johnson (1995) use the regression tree method of Brieman et al. (1984) to estimate a version of the human-capital-augmented Solow growth model introduced by Mankiw, Romer, and Weil (1992) that allows (endogenously determined) subgroups of countries to obey different growth equations. They show that such a model fits the Mankiw, Romer, and Weil (1992) data better than the linear model used by those authors. This finding is consistent with the view that there are multiple basins of attraction in the process describing the evolution of output per capita. In this case, initial conditions (the determinants of which basin each country belongs to) rather than just cross-country variation in the variables that determine the steady state in the Solow model, may be necessary to explain long-run differences in cross-country growth behavior.

5.2 *Nonlinear Models*

A desire to model the potential deviations from β -convergence and improve upon tests of it based on equation (3) has given rise to a variety of nonlinear models as they admit

the economically interesting possibilities of divergence and club convergence. The possibility that different subsamples might display behavior consistent with different long-run outcomes had been considered as early as Baumol (1986) who divides his sample into the industrialized countries, the centrally planned economies, and the rest (depending on their status in 1950) to explore the possibility of nonlinearities in the relationship between initial income per capita and subsequent growth and so the implied existence of more than one "convergence club." Other examples of exogenous sample splitting include Baumol and Wolff (1988), Grier and Tullock (1989), and Dowrick (1992), all of whom find evidence consistent heterogeneous behavior across different subsamples in their data.

Many other papers have considered a wide variety of nonlinearities using an equally wide variety of statistical models. We consider them in three not entirely distinct groups: (i) models that cluster the data into groups of countries that obey common models; (ii) models with smoothly varying parameters, so that while a continuous equation describes the data, the equation is nonlinear; and (iii) methods that consider the shape and evolution of the cross-country distribution of per capita income.

5.2.1 *Models with Smoothly Varying Parameters*

This approach to parameter heterogeneity permits the parameters of the growth regression to vary smoothly across countries. Again, Baumol and Wolff (1988) provide an early example by allowing the coefficient on initial income to vary with its square and find that the negative coefficient required by the catch-up hypothesis is evident only for higher values of initial income. Chatterji (1992) estimates a cubic relationship between (the log of) income per capita (relative to its level in the United States) in 1985 and that in 1960,

and concludes that the implied nonlinear difference equation for relative income has two basins of attraction with countries belonging to one or the other (and hence displaying divergent long-run behavior) depending on whether or not their initial income per capita is greater or less than about one-sixth of the US value.

Liu and Stengos (1999) estimate a semi-parametric additive partially linear growth regression that allows the coefficients on a measure of human capital accumulation and initial income to vary smoothly with the levels of the respective variables and, like Baumol and Wolff (1988), they find that the coefficient on initial income is negative only for higher values of that variable.

Arguing that the “homogeneity assumptions” imposed on growth regressions were not envisaged by the original crafters of the theory on which they are based, Durlauf et al. (2001) extend the approach of Liu and Stengos (1999) to allow all of the coefficients of the growth regression implied by the augmented Solow model of Mankiw et al. (1992) to vary with initial income, so that while the model is locally (i.e., for any value of initial income) Solow, globally it (potentially) exhibits parameter heterogeneity. This extension confirms the finding that the coefficient on initial income is negative only for higher values of initial income, and also reveals strong evidence of substantial heterogeneity in the parameters relating the Solow variables to economic growth. Henderson (2010) proposes estimation of the density of the coefficient on initial income in a nonparametric growth regression and finds a multimodal density with a mode centered on negative values of β and two others centered on positive values, implying β -convergence for some countries (identified as the OECD), but not for most. The evidence on heterogeneity is further strengthened by Kourtellis (2011), who allows the coefficients to depend on initial literacy and initial life expectancy.

Related work includes Banerjee and Duflo (2003); Ketteni, Mamuneas, and Stengos (2007); Stengos et al. (2007); Minier (2007a, 2007b); and Sirimaneetham and Temple (2009), all of which find evidence of nonlinear relationships between growth and its determinants.

Nonparametric estimation methods provide a way to investigate the existence and nature of nonlinearities without the need to specify the variables that govern parameter heterogeneity, as is the case for the partially linear regression model. Maasoumi et al. (2007) use a nonparametric local linear estimator to estimate growth regressions with the “Solow variables” on the right-hand side for both OECD and non-OECD samples. They find considerable variation in the relationships between growth and its determinants across the two samples, as well as in deviations from linearity within each sample. Owen, Videras, and Davis (2009) estimate a finite mixture model for the conditional distribution of growth rates and conclude that the growth process is characterized by multiple regimes with institutional quality being an important determinant of which countries obey which regimes. Henderson et al. (2012) employ nonparametric estimation of regression functions in the presence of “irrelevant” regressors with results that underscore the importance of nonlinearities in growth regressions.

5.2.2 *Distributional Models*

Critical of the fact that the existing methods of studying convergence considered only a few moments of the distribution of output per capita, Quah (1993a, 1993b, 1996a, 1996b, 1996c, 1997) pioneered the “distribution dynamics” approach to studying the role of nonlinearities in economic growth. These methods originally employed Markov chains to study the evolution of the cross-country distribution of income per capita, but the required discretization of the

state space of a continuous random variable (typically per capita income) changes the probabilistic properties of the data and more recent applications employ almost exclusively continuous state space methods. This body of research has found substantial evidence of convergence clubs when data from a large group of countries are used.²⁴ Quah's work demonstrated the existence of "twin peaks" in the long-run cross-country income distribution—two modes, indicative of two basins of attraction in the growth process. Henderson, Parmeter, and Russell (2008) confirm the multimodality of the cross-country distribution of per capita output using a variety of measures of the concept and a variety of statistical tests.²⁵

More recent work has suggested the existence of more than two basins of attraction.²⁶

²⁴A selection of the research applying and extending Quah's ideas, in addition to the research cited in Durlauf, Johnson, and Temple (2005), includes Fiaschi and Lavezzi (2003a, b and 2007a, b), Fotopoulos (2006, 2008), Maasoumi, Racine, and Stengos (2007), Fischer and Stumpner (2008), and Bandyopadhyay (2011).

²⁵Several authors, including Beaudry, Collard, and Green (2005), Johnson (2005), Feyrer (2008), Barseghyan and DiCecio (2011), and Badunenko, Henderson, and Russell (2013) have sought the proximate causes of the shape of the cross-country distribution of per capita and its changes, but there is no apparent consensus regarding the relative importance of cross-country variation in total factor productivity (TFP) or physical or human capital accumulation. Battisti, di Vaio, and Zeira (2016) extend the conditional β -convergence approach by modeling the technology adoption of each country and relaxing the assumption that all countries follow the global technology frontier. While they find that output per worker in each country converges to its own productivity path, they conclude that many countries are diverging from the global technology frontier which, they suggest, is an important source of the lack of convergence in cross-country income levels. See also Comin and Mestieri (2018).

²⁶Krause (2017) uses the critical bandwidth (the largest bandwidth allowing for a finding of bimodality) as measure of the coalescence of countries around the peaks in the cross-country distribution of per capita income. Analyzing data from 1970 to 2011, she finds a tendency for the countries of the world to converge to two groups, which has weakened since 2000. While this approach can be criticized for conflating peaks and convergence clubs, it is interesting to note that the author attributes this "de-clubbing" to the rapid growth recently observed in some low income

Pittau, Zelli, and Johnson (2010) point out that multimodality is neither necessary nor sufficient for the existence of convergence clubs and, following Paap and van Dijk (1998), Tsionas (2000), Pittau (2005), and Pittau and Zelli (2006a, b), estimate a finite mixture model of the cross-country income distribution. They find that it has three (data-selected) components, a result confirmed by Battisti and Parmeter (2013) who generalize the approach of Pittau, Zelli, and Johnson (2010) and exploit the panel nature of the data set.²⁷

As they find little movement between the three components, Pittau, Zelli, and Johnson (2010) interpret them as evidence of three basins of attraction in the growth process and note that the gap between the implied middle-income and high-income groups has become wider and more pronounced since the early 1970s. Anderson, Pittau, and Zelli (2016) take a related approach and attribute the recent growth in the gap between the rich and other countries to the tendency for many middle income countries to fall back into the poor group. Pittau, Zelli, and Johnson (2010) also present evidence of an increase in the gap between the typical poor country and the typical rich country. This is partly due to an increase in the gap between the mean incomes of these two components but, more importantly, to a decrease in the dispersion of the two groups of countries around their respective component means. Together, these two changes drive a documented rise in polarization in the cross-country income distribution.

Epstein, Howlett, and Schulze (2003) apply Quah's distribution dynamics methods to data from seventeen OECD countries covering 1870–1992. While the

countries discussed in section 2 above. See also Anderson et al. (2012).

²⁷See also Vollmer, Holzmann, and Schwaiger (2013) and Pittau, Zelli, and Massari (2016).

pre-1914 and 1914–50 periods appear to be characterized by persistence within the cross-country distribution of per capita incomes, mobility and some convergence is observed in the post-1950 period using data for the larger group of 24 countries available for that period. However, this gives way to some divergence beginning in the early 1970s. The contrast between the two eras of increased economic openness, 1870–1914 and post-1950, is affirmed by Di Vaio and Enflo (2011), who estimate a mixture model for the growth rate of per capita GDP using data from 1870–2003 for a larger group of countries. Their approach specifies a standard growth regression for each of the (data-determined) regimes with regime membership being determined stochastically, and the estimated coefficient on initial income being used to perform a standard (within-regime) convergence test.

These papers can all be considered as part of the literature that searches for evidence of multiple basins of attraction using various clustering algorithms. Earlier work of that type includes Desdoigts (1999) and Kourtellos (2002) who use projection pursuit methods and find evidence of multiple steady states although, apart from the OECD countries, those in the former paper are closely linked to geography. Canova (2004) uses initial per capita income to order the OECD countries and finds that there are two clusters with an “economically large” implied difference in the long-run incomes and little mobility between them.

A common aspect of much of the literature considered so far in this section is the use of per capita income as the variable defining basins of attraction on the growth process. As noted above, the case can be made that per capita income is not a sufficient statistic for a country’s initial conditions, inviting the use of other variables in defining convergence clubs. For example, Durlauf and Johnson (1995) found that literacy rates as

well as initial per capita income were useful in identifying convergence clubs. More recently, Tan (2010) uses a regression-tree method that generalizes that used in Durlauf and Johnson (1995) and attributes a dominant role to similarities in institutional quality and ethnic fractionalization in identifying countries with similar long-run behavior. Importantly, he finds no such role for geographic factors. Battisti and Parmeter (2013) use mixture models to model the joint distribution of output per capita and its proximate determinants—physical and human capital and TFP—and find evidence of multiple clusters. Fiaschi, Lavezzi, and Parenti (2018) consider a large number and wide variety of possible deep determinants of variation in output per capita as candidates for defining growth clusters. Using a method based on the Akaike information criterion (AIC), they find that initial conditions define three clusters according to life expectancy in 1960 and the share of Catholics in the population in 1965. The former is considered a measure of human capital while the latter is considered to be a measure of culture.

Vollmer, Holzmann, and Schwaiger (2013) argue for the emergence of three “human development clubs” during the 1990s using a finite mixture model of the joint distribution of income per capita, educational attainment, and life expectancy, which they also conclude has three components. They note the coherence between these findings of three components and the recent literature discussing the “middle-income trap” which, as Kharas and Kohli (2011) discuss, is the name given to the phenomenon wherein many countries experience abrupt slowdowns following periods of rapid growth. These countries become trapped between the low-income countries, with whom they are unable to compete because their wage structure is too high, and the advanced countries, with whom they are unable to compete because their technology structure is not suf-

ficiently advanced. As Vollmer, Holzmann, and Schwaiger (2013) note, the middle-income trap may well prove to be a transitory stage of development—an example of the identification problem discussed by Durlauf, Johnson, and Temple (2005, pp. 622–3). El-Gamal and Ryu (2013) document the appearance, disappearance, and reappearance of a “stochastically stable” middle-income group over the 1960–2009 period. This could reflect the lack of stationarity in the transition dynamics that they observe, complex nonlinearities, or a dynamic process of higher than first order. The dynamic process followed by the cross-country distribution of per capita income has only ever being modeled as a first-order process, so the implied misspecification if the process is of higher order could manifest as a lack of stationary.

In studying σ -convergence, another body of research has focused on the dispersion of the cross-country distribution of per capita incomes.²⁸ The most commonly used measures of dispersion are the standard deviation and the coefficient of variation of the log of cross country income (see, e.g., Barro and Sala-i-Martin 1991, Ben-David 1993, and Slaughter 2001). However, other indices exist with interesting properties (see Cowell 1995), including spatial properties that affect measures of σ -convergence (Bode and Rey 2006, Egger and Pfaffermayr 2009). It is important to note that σ -convergence tests lack power against the club convergence alternative in much the same way as do β -convergence, as there is no reason why the dispersion of the cross-country distribution of per capita income cannot decline as when countries lie in two or more basins of attraction.

The σ -convergence concept has seen a host of recent applications at the regional

level but far fewer at the global level. Barro (2012) reports a tendency for the standard deviation of the logs of per capita GDP and consumption to decline since the mid-1970s for a group of mostly industrialized countries. Madsen and Timol (2011) use the tests of Lichtenberg (1994) and Carree and Klomp (1997) to examine the σ -convergence hypothesis using data on labor productivity in the manufacturing sectors of nineteen OECD countries since 1870, and conclude that σ -convergence (and also β -convergence) occurred, a result that is subject to De Long’s (1988) critique.²⁹ Rodrik (2013) finds σ -convergence in labor productivity in the manufacturing sector at the two-digit level for a smaller sample of countries. The time-series convergence tests that allow for transition dynamics formulated by Phillips and Sul (2007a, 2007b, 2009) can be considered as tests of σ -convergence and we discuss these below.

The conclusion that we take away from the research on nonlinear models of growth is that, once the alternative hypothesis of club convergence is explicitly considered, there is strong evidence for a rejection of the view that initial conditions do not matter for long-run outcomes in favor of the view that they do and that, as a result, the cross-country distribution of per capita income exhibits characteristics consistent with the existence of two or more convergence clubs. We are, however, mindful of an important identification caveat emphasized by Durlauf and Johnson (1995) that needs to be attached to convergence tests: given the finite spans of data available, it is impossible to unequivocally distinguish between a model in which there are multiple steady states and a model in which countries transition through different stages of development before reaching a common steady state.

²⁸ Durlauf, Johnson, and Temple (2009) discuss some of the econometric issues that arise in the application of the σ -convergence concept.

²⁹ See also Bernard and Jones (1996) and Wu (2009).

5.3 Time Series Models

So far the evidence we presented relates to the strand of literature that focuses on country cross-sections or panels. Next we turn attention to covering time series approaches that have recently been quite prevalent notwithstanding transitional data constraints.

5.3.1 Structural Breaks

Perron (1989) shows that trend or structural breaks reduce the power of unit-root tests and many researchers have found that permitting structural breaks in the form of mean or trend shifts, either exogenously or endogenously determined, makes it more likely that the null hypothesis of a unit root will be rejected.³⁰ That is, as statistical matter, the inclusion of trend or structural breaks that are often found to coincide with the Great Depression or the Second World War, or restricting the data used to that from the postwar period, tends to produce test results more favorable to the convergence hypothesis.³¹

It is not clear, however, how to interpret these results, in part because the interpretation of the breaks is unclear. Do the breaks represent large exogenous shocks? Or are they the manifestation of the persistent effects of shocks? Or do they reflect un-modeled nonlinearities or transition dynamics in the growth process? Answering these questions would require a model of the growth process that allows a role for such shocks to inform the empirical work.

³⁰ Stock (1994) surveys the literature on the relationship between structural breaks and unit-root tests.

³¹ A selection of the research studying the role of structural breaks in convergence tests includes Carlino and Mills (1993), Oxley and Greasley (1995), Loewy and Papell (1996), Greasley and Oxley (1997), Li and Papell (1999), Strazicich, Lee, and Day (2004), Dawson and Sen (2007), Dawson and Strazicich (2010), Costantini and Sen (2012), King and Ramlogan-Dobson (2014), and Ghoshray and Khan (2015).

Moreover, the implications of the general finding that allowing trend breaks reduces the tendency to not reject the unit root null hypothesis for the convergence hypothesis itself are also not clear. If the process for $y_{i,t} - y_{j,t}$ is found to be stationary once a structural break is permitted, a conclusion of convergence is then conditional on the occurrence of the break.

5.3.2 Long-memory Approaches

Michelacci and Zaffaroni (2000) argue that the early time series tests of convergence are flawed because $y_{i,t}$ obeys a long-memory process which, following Granger (1980), they motivate by considering a Solow growth model for an economy with heterogeneous sectors. The original time series tests of convergence assume that $y_{i,t}$ is either $I(0)$ or $I(1)$, perhaps around a deterministic trend. Fractionally differenced, $I(d)$, models allow d , the exponent on the first difference operator in autoregressive integrated moving average (ARIMA) representation of $y_{i,t}$, to differ from the polar cases of zero and one, which imply, respectively, a rapid decay in the effects of shocks or none at all. For $0 < d < 1$, $y_{i,t}$ is mean reverting but it exhibits “long memory” and is more persistent than a standard autoregressive moving average (ARMA) process.³²

While they do not explicitly test for convergence in a long-memory framework, Michelacci and Zaffaroni (2000) do find evi-

³² Specifically, if x_t is a zero mean stochastic process with the representation $A(L)(1-L)^d x_t = B(L)u_t$, where $A(L)$ and $B(L)$ are lag polynomials with roots outside the unit circle and u_t is a white-noise process, then d can take any real value rather than being constrained to 0 or 1. For $d = 0$, x_t is a standard ARMA process with short memory, but for $0 < d < 0.5$, x_t is a covariance-stationary long-memory process because its autocorrelation function decays hyperbolically rather than geometrically. For $0.5 \leq d < 1$, x_t is also long-memory process that, while mean reverting, is no longer covariance stationary; and for $d > 1$, x_t is explosive. See Baillie (1996) for a survey of fractional integration and long-memory processes.

dence of long-memory behavior in output data from the OECD countries in the form of estimates of d between 0.5 and 1.0, which they claim is both consistent with the 2 percent rate of convergence commonly found in earlier cross-country analyses and damaging to the reliability of unit-root tests of convergence.

Silverberg and Verspagen (2003) offer a motivation for long-memory representations of output by suggesting that evolutionary models of economic behavior can endogenously generate the cross-sectional heterogeneity that drives Granger's (1980) aggregation rationale. They employ a variety of empirical approaches designed to overcome many of the criticisms of the Michelacci and Zaffaroni (2000) data analysis, and their conclusions about the suitability of long-memory representations of output are far more agnostic. One important difference between the two papers is how the trend is modeled. Michelacci and Zaffaroni (2000) assume a linear trend in the log of per capita incomes while Silverberg and Verspagen (2003) first-difference their data prior to analysis.

Mean reversion in per capita incomes implies that definition (4) above could still be satisfied, but standard tests of the unit root null in the $y_{i,t} - y_{j,t}$ process will have low power against a long-memory alternative because of the relatively slow decay of the effects of shocks under the alternative hypothesis. Cunado, Gil-Alana, and Pérez de Gracia (2006) test for convergence by examining the (possibly fractional) order of integration of the deviation of log per capita income in fourteen OECD countries from that in the United States: $y_{i,t} - y_{US,t}$. Using a data set beginning in the late nineteenth century, they are unable to reject the hypothesis of a unit root for almost all countries, but restricting the analysis to the postwar period leads to confidence intervals for d the lie in (0.5, 1), consistent with con-

vergence in the sense of definition (4) above, for almost all of the fourteen countries.³³ Silverberg and Verspagen (1999) report similar results. Dufrénot et al. (2011) investigate the behavior of $y_{i,t} - y_{j,t}$, where country j is a regional benchmark country (selected as the country in the region with the highest per capita GDP at the end of the sample) using postwar data on a group of ninety-eight developing countries. They find considerable variation in the time-series properties of $y_{i,t} - y_{j,t}$ across countries within regions and across regions. Cunado, Gil-Alana, and Pérez de Gracia (2006) and Dufrénot, Mignon, and Naccache (2012) include a time trend in their time-series representations for $y_{i,t} - y_{US,t}$ and $y_{i,t} - y_{j,t}$ which, as we discuss below, permits some transition dynamics but apparently does not increase the power of their tests. For example, Cunado, Gil-Alana, and Pérez de Gracia's (2006) confidence intervals for d for the entire sample are little changed by the introduction of intercept and trend terms while those for the postwar data become wider.

To avoid the problem of choosing a benchmark country, Stengos and Yazgan (2014) follow Peseran (2007) in considering pairwise convergence by examining the value of d for all possible values of $y_{i,t} - y_{j,t}$ for 139 countries in the postwar period. They find $d > 0.5$ but cite a "lack of power" in concluding that the distinction between $I(0)$ and $I(1)$ processes adequately describes the behavior of log per capita output levels. That is, they conclude against convergence despite allowing for smooth structural breaks by including a Fourier function of time in the representation for $y_{i,t} - y_{j,t}$. Stengos et al. (2016) take a multivariate approach to the estimation of d and find stronger evidence of mean rever-

³³That the countries for which the unit root cannot be rejected in the postwar period are Austria, Germany, and Japan may underscore the importance of transition dynamics as we discuss below.

sion (i.e., $0.5 < d < 1$) and hence convergence than Stengos and Yazgan (2014).

While the finding that data from the postwar period are more likely to be favorable to the convergence hypothesis than data over from the first half of the twentieth century is noteworthy, one difficulty in interpreting these results is the inclusion of trends (Cunado, Gil-Alana, and Pérez de Gracia 2006, Dufrénot et al. 2011) and trend breaks (Stengos and Yazgan 2014, Kourtellis, Stengos, and Tan 2016) or first differences (Silverberg and Verspagen 1999) in the representations for the income gaps, which have an undocumented effect on the results and raise the same sorts of issues as the trend-break literature discussed above. Further complicating matters, Diebold and Inoue (2001) show that, as a matter of theory, apparent long memory can be the manifestation of some types of structural change in $I(0)$ or $I(1)$ processes and Granger and Hyung (2004) show that distinguishing between fractionally integrated processes and processes with occasional breaks can be difficult.

Another concern with this literature is the lack of strong theoretical reasons to believe that per capita outputs obey fractionally integrated processes. Michelacci and Zaffaroni (2000) and Silverberg and Verspagen (2003) suggest reasons why fractionally integrated processes might arise from growth models, whereas Lau (1999) shows that integrated and cointegrated processes arise naturally under in a wide class of growth models.

5.3.3 Transition Dynamics

As Bernard and Durlauf acknowledge, tests based on the co-integration of cross-country income levels are more appropriately regarded as tests that convergence has occurred than tests that convergence is occurring. It is possible, for example, that the expected difference in two countries' log per capita incomes at any time

$t + T$, $E(y_{i,t+T} - y_{j,t+T} | F_t)$ includes a deterministic term $\mu_{t+T} \neq 0$ where $\lim_{T \rightarrow \infty} \mu_{t+T} = 0$ so that convergence definition (4) is satisfied despite a contemporary nonzero expected difference in incomes because at least one of the countries has not yet reached its steady state. Data generated by such a process may be less likely to produce a rejection of a unit root null hypothesis in the $y_{i,t} - y_{j,t}$ process if $\mu_{t+1} \approx \mu_t$ because convergence is slow as the difference in incomes will then have a highly persistent component.

Oxley and Greasley (1995) add a time trend to the representation for $y_{i,t} - y_{j,t}$ which, omitting higher order lagged terms, they write as $y_{i,t} - y_{j,t} = \mu + \alpha(y_{i,t-1} - y_{j,t-1}) + \beta t + \varepsilon_t$. They define long-run convergence as $\alpha < 1$ and $\beta = 0$ (although to satisfy definition (4), $\mu = 0$ is also necessary), while $\alpha < 1$ and $\beta \neq 0$ is defined as "catching up." Using data for the US, UK, and Australian economies from the late nineteenth to the late twentieth centuries, and allowing for the possibility of discontinuities in the trend, they find a role for the catch-up term for the US/UK and US/Australia pairs, but not for the UK/Australia pair, which they conclude have converged. Chong et al. (2008) estimate a nonlinear version of this model using postwar data on $y_{i,t} - y_{US,t}$ where i is an OECD country. For those twelve countries with apparently nonlinear income gaps to the United States, the unit-root null is not rejected in eight cases, suggesting that transition dynamics per se are not the entire reason for the rejections of convergence found in time series tests. King and Ramlogan-Dobson (2011) find that allowing for breaks in the trend function that captures the transition dynamics weakens the statistical case against convergence found using this approach.

A related approach is taken by Nahar and Inder (2002), who model the deviation of per capita output in country i from that in the United States at time t , $y_{i,t} - y_{US,t}$, as a

polynomial in t . They test the hypothesis that the derivative of the polynomial with respect to t is zero against that alternative that it is positive which implies that country i is catching up to the United States. Bentzen (2005) modifies this approach to allow the rate of convergence to vary over the sample period. While this approach does highlight the possible richness of transition paths for different countries, it specifies the non-convergence null hypothesis as the overly restrictive requirement that $y_{i,t} - y_{US,t}$ is an i.i.d. random variable, so it is not clear how this test performs when the non-convergence of $y_{i,t} - y_{US,t}$ causes it to follow an integrated process. Datta's (2003) time-varying parameter approach to modeling transition dynamics also finds evidence of catching up using postwar data from the OECD countries, but is subject to a similar criticism.

Phillips and Sul (2007a, 2007b, 2009) present a time-series test of convergence that allows for substantial cross-country heterogeneity in the transition dynamics. They write $y_{i,t} = b_{it}\mu_t$ where μ_t is the hypothesized steady-state growth path common to all counties and b_{it} describes the transition path of economy i to the steady-state growth path. Convergence is said to occur if $\lim_{t \rightarrow \infty} h_{it} = 1$ for all i where $h_{it} = n y_{i,t} / \sum_{i=1}^n y_{i,t} = n b_{it} / \sum_{i=1}^n b_{it}$. Phillips and Sul then propose testing for convergence by specifying a model of the transition path that yields a test based on a conventional one-sided t test of the hypothesis $\gamma = 0$ (no convergence) against the alternative $\gamma > 0$ (convergence) in the regression $\log(H_1/H_t) - 2 \log(\log t) = a + \gamma \log t + u_t$, where $H_t = n^{-1} \sum_{i=1}^n (h_{it} - 1)^2$.³⁴ They show that γ is twice the speed of convergence and that, when μ_t follows a random walk with drift or trend stationary process, $0 < \gamma < 2$

implies convergence in growth rates (conditional convergence) while $2 \leq \gamma$ implies convergence in levels (absolute convergence), that is in the sense of definition (4). Phillips and Sul (2009) apply this test to data from the United States, the Western OECD countries, and 152 countries from the PWT and, consistent with much of the literature, find evidence of convergence in growth rates, but not in levels, for the US states (1929–98) and the Western OECD countries (1870–2001 and 1940–2001, but not 1870–1929 nor 1911–70) nor for the PWT countries.

5.3.4 Clustering Approaches

There have also been some applications of clustering approaches in the time-series context to check for club convergence. Phillips and Sul (2007a) develop a clustering algorithm designed to divide the countries in a data set into groups less heterogeneous than the sample as a whole and then to test for convergence within groups using the method described above. Having rejected convergence for the 1970–2003 sample of 152 PWT countries, Phillips and Sul (2009) apply this approach to that data set and conclude that it can be split into four convergence clubs (three of which exhibit growth rate or conditional convergence) and a small group of diverging countries.³⁵ Finding between two and five convergence clubs is common and consistent with the results of those who've studied the evolution of the cross-country distribution of per capita income such as Pittau, Zelli, and Johnson (2010).

By contrast, Beylunioglu, Stengos, and Yazgan (2016) extend Peseran's (2007) pair-wise time series approach using a

³⁴Observe that $n^{-1} \sum_{i=1}^n h_{it} = 1$ so that H_t is the sample variance of h_{it} implying that the Phillips and Sul approach may also be considered as a test of σ -convergence.

³⁵The memberships of the clubs different from those found by Durlauf and Johnson (1995) using a regression tree algorithm on data from ninety-eight countries for 1960–85 from an antecedent to the PWT, but it is not clear whether these differences can be attributed to the different number of countries, the different time periods, or the different clustering algorithm.

“maximal clique” approach that, like that of Hobijn and Frances (2000), is also based on time-series convergence concepts, and finds a large number of very small clusters that the authors call “convergence clubs.” While these results are evidence against convergence the large number of clusters found makes the convergence clubs interpretation difficult to sustain. Moreover, it is likely that the groups found reflect countries between which convergence has occurred, rather than those between which convergence is occurring. A test along these lines that also took into account transition dynamics would, no doubt, produce fewer groups as do, for example, Phillips and Sul (2008).

Our reading of the recent time-series tests of the convergence hypothesis is broadly in accord with that of the nonlinear cross-section studies of the hypothesis discussed above. While there may be some weak support for the proposition that gaps between per capita income levels have fallen in some groups of countries in the postwar period, the interpretation of whatever support that there is for the convergence hypothesis is fraught with difficulty because of the lack of consensus about how to treat the trends in the data and, in particular, for the meaning for convergence of the structural and trend breaks that are often found. Importantly, as we observed above, once the alternative hypothesis of convergence clubs is considered by use of a clustering algorithm, there is evidence in favor of that view that is consistent with that found using other clustering approaches.

5.4 Cross-Individual Distribution of Income

Beyond the analysis of the rich literature presented above that is based on country convergence—unit of analysis being the country—recent work attempts to assess global inequality by considering the distri-

bution of incomes of individual households from different countries around the globe. The advantage of a global approach lies in the detail with which we can observe and analyze the effects of globalization in different segments of the global income distribution. These studies combine micro and macro data coming from hundreds of household surveys from over one hundred countries in the world, covering more than 90 percent of the world population and income. Advocates of this approach argue that using individuals or households, rather than countries, as units of analysis is more useful if one is concerned about human welfare because different countries have different population sizes. As stated by Sala-i-Martin (2006), “After all, there is no reason to down-weight the well-being of a Chinese peasant relative to a Senegalese farmer just because the population in China is larger than that of Senegal.”

Excellent examples on world income distribution and global inequality include Milanovic’s early paper (1997) that lays down many of the issues discussed by the subsequent literature, Bourguignon and Morrisson (2002), Atkinson and Brandolini (2004), Sala-i-Martin (2009), Deaton (2010), Pinkovskiy and Sala-i-Martin (2014), Atkinson (2015), Bourguignon (2015), and Lakner and Milanovic (2015).

Branko Milanovic’s recent book *Global Inequality: A New Approach for the Age of Globalization* (2016a) may well summarize the most current thinking on global inequality. Milanovic’s book, like Bourguignon’s (*The Globalisation of Inequality*, 2015), concludes with one key fact: while inequality is rising within most countries, notably the high-income ones, global inequality of incomes, though huge, has been falling, particularly since 2000. This comes as no surprise given the millions of Chinese lifted from poverty during China’s historic growth acceleration of the last three decades

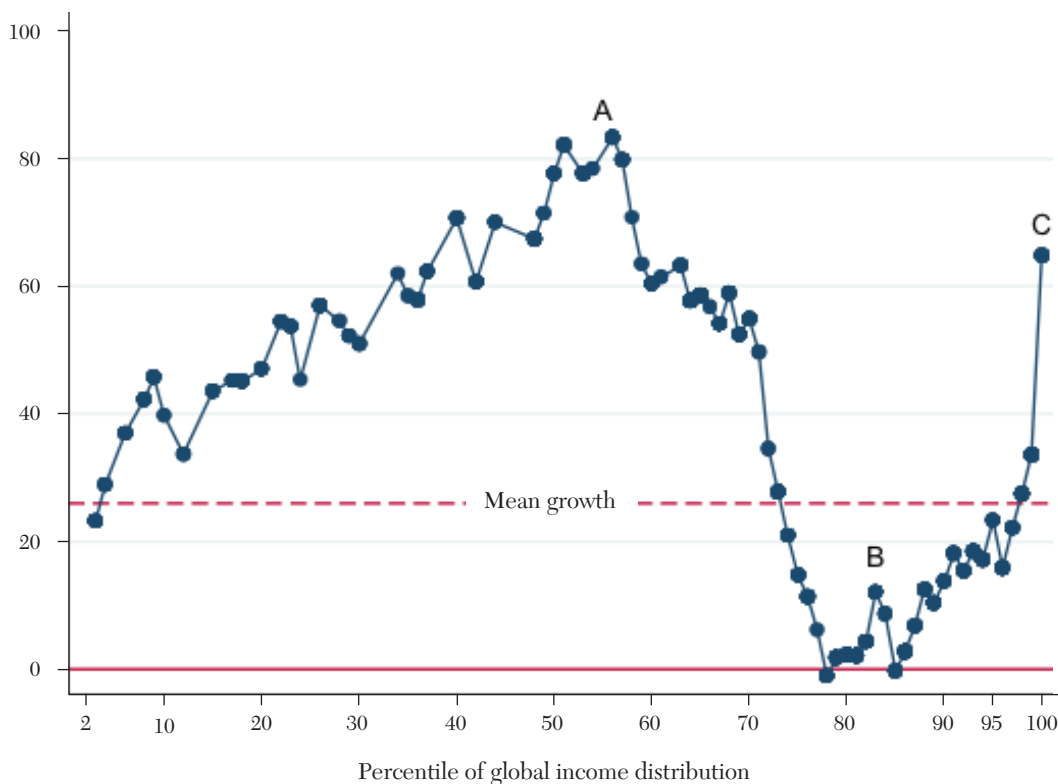


Figure 8. Cumulative Real per Capita Income Growth at Various Percentiles of the Global Income Distribution (1988–2011)

Source: Milanovic (2016a). Reprinted with permission from Harvard University Press.

discussed previously. Yet many economists argue that this positive trend might not continue, once global favorable conditions slow down and as China's wages rise further. Prospects would depend on whether China's economic progress can be replicated and African or other Asian nations (e.g., India, Bangladesh, Vietnam) can follow suit.

Another interesting fact from Milanovic's work is illustrated in figure 8 showing the proportional rise in real per capita incomes across the world income distribution between 1988 to 2011. Income is measured in 2005 international dollars and individuals

are ranked by their real household per capita income. On the one hand, points A and C on the cumulative distribution plot reflect large real income gains made by individuals around the global median, and by those who are part of the global top 1 percent, respectively. On the other hand, those at the bottom part of the distribution have done relatively poorly, and those between the eightieth and ninety-fifth percentiles have seen stagnant real incomes (point B). In the words of Milanovic (2016b), "The people around the global median are, however, still relatively poor by Western standards. This

emerging ‘global middle class’ is composed of individuals with household per capita incomes of between 5 and 15 international dollars per day. The contrast between the unambiguous success of people at point A and the relative failure of people at point B allows us to look at the effects of globalization more broadly.”

Angus Deaton in his book, *The Great Escape* (2013), offers a novel insight on global inequality. He argues that the world is a better place than it used to be, but that the enormous progress that resulted in catching up of some countries was also responsible for opening up gaps and setting the stage for today’s disproportionately unequal world. In the words of Deaton, “Rapid economic growth in many countries has delivered hundreds of millions of people from destitution. . . . As always, progress has not been even; some of the most rapidly growing countries have narrowed the gap with the rich countries, but their progress has opened up new gaps between them and the countries left behind. Once-poor countries in Asia have moved into the middle, leaving chasms between them and many countries in Africa” (p. 218).

6. Conclusion

In its simplest form, convergence suggests that poor countries have the propensity to grow faster than the rich, so to eventually catch up to them. The idea of convergence has its formal origins in Solow (1956), but its empirical treatment really begins in the mid-1980s being further motivated by the modern growth theory and empirics; it remains today a perennial research topic although only, perhaps, under the new lamp-post of global inequality.

The voluminous literature that has emerged seeking answers to the convergence question is a testimony to the interest that the hypothesis has generated—a crude inter-

net search reveals that over the last thirty years there have been thousands of papers written on the subject; in addition, four of the most influential papers in the growth literature focus on convergence and account for almost 11,000 citations (according to Google scholar).³⁶ Despite the concept of the long-run irrelevance of initial conditions being straightforward, empirically testing convergence and understanding its mechanics proved quite elusive. As shown in this survey, convergence is hard to pin down, first, because the concept can be operationalized in many ways and second, because econometric approaches and data measurement issues remain a challenge in empirical tests of convergence. Ultimately, understanding convergence would get us closer to understanding the process of economic growth—a truly humbling endeavor. So, what is the evidence on convergence and what have we learned from it thus far?

With the exception of some early studies that have been criticized extensively due to econometric problems, there is a broad consensus of no evidence supporting *absolute convergence* in cross-country per capita incomes—that is poor countries do not seem to be unconditionally catching up to rich ones. The only glimmer of hope for this hypothesis is provided by Rodrik’s (2013) finding of unconditional convergence in particular *manufacturing* industries. Of course, whether this result will hold up to the scrutiny of subsequent researchers, armed with more sophisticated methods and better data, is unknown.

A more definitive conclusion from this survey is that the process of growth and so, potentially, of convergence, is not smooth but rather start and stop, and is characterized by significant country heterogeneity. Such realization must not come as a surprise given that

³⁶As of October 24, 2016, Barro and Sala-i-Martin (1991) 4,262 citations; Baumol (1986) 3,939; De Long (1988) 1,464; and Bernard and Durlauf (1996) 1,188.

the notion of convergence is only a theoretical construction that characterizes part of the broader dynamic growth process across countries. Under such a framework, it is then possible that several mechanisms of divergence and convergence are concurrently at work across countries in different stages of their development process. For example, the divergence process may tend to dominate the convergence process at early stages of economic growth, while the reverse could hold true for later stages (Steger 2006 and International Monetary Fund 2017 make a similar observation). In this regard, exploring those mechanisms that induce convergence and those that result in non-convergence or divergence would be central to understanding such phenomena. Also, focusing attention to potential mechanisms that determine growth dynamics (linear and nonlinear) or the distribution of global inequality, as it is done in recent studies, may be more tenable and informative than seeking evidence on convergence.

Our reading of the evidence, then, is that recent optimism in favor of rapid and sustainable convergence is unfounded. The last two decades of an unprecedented wave of growth in many LICs and emerging markets led to many analysts claiming prematurely, in our view, success with slogans such as “lions on the move,” “the next convergence,” and “no shortage of economic growth in Africa” (Roxburgh et al. 2010; Spence 2011; Economist 2013, respectively). Many observers are led to believe that “this time is different.” We have come to the conclusion that with the exception of a few countries in Asia that exhibited transformational growth, most of the economic achievements in developing economies have been the result of removing inefficiencies, especially in governance and in political institutions. But as is now well known, these are merely one-off *level* effects that, while not unimportant, and in fact necessary in the process of development,

nonetheless do not stimulate ongoing economic growth.

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